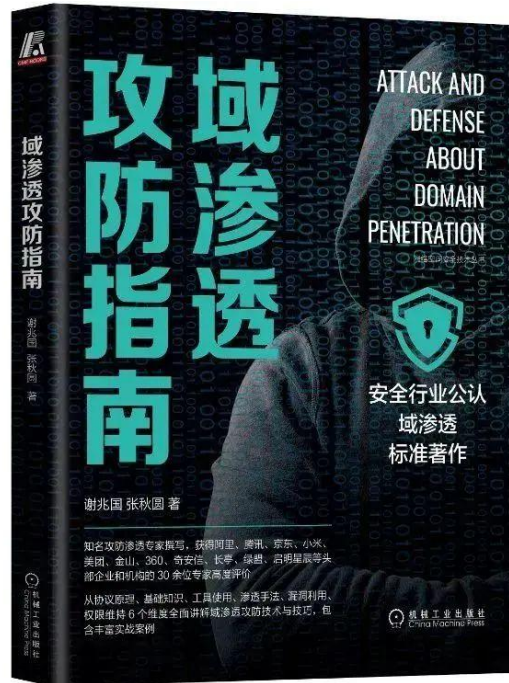


## 目录分区 Directory Partitions

本文部分节选于《域渗透攻防指南》，购买请长按如下图片扫码



小谢1997521 为你推荐

【限量签名版】《域渗透攻防指南》

¥99 ¥129



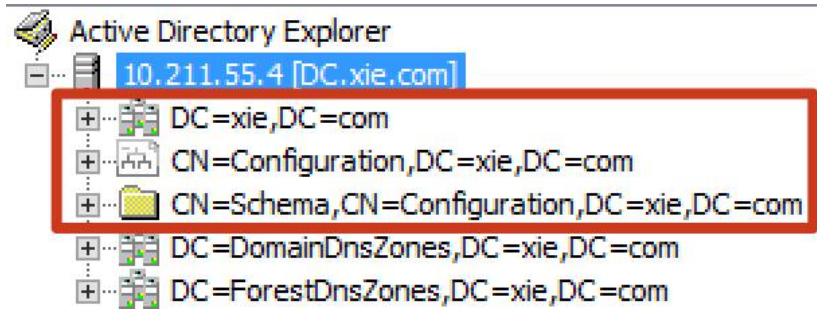
本商品由 放心购 提供保障

谢公子学安全

由活动目录域服务 ADDS 控制的域林中的每个域控制器都包含目录分区 Directory Partitions。目录分区 Directory Partitions 也被称为命名上下文(Naming Context，简称 NC)。目录分区 Directory Partitions 是具有独立的复制范围和调度数据的整个目录的连续部分。默认情况下，企业的活动目录域服务包含以下目录分区：

- 域目录分区(Domain Directory Partition): 每一个域各有一个域目录分区, 域目录分区包含与本地域相关联的目录对象, 如用户和计算机等。一个域可以有多个域控制器, 一个林可以有多个域。每个域控制器为其本地域存储域目录分区的完整副本, 但不存储其他域的域目录分区的副本。如图中的 DC=xie,DC=com 就是域目录分区。
- 配置目录分区(Configuration Directory Partition): 配置目录分区包含复制拓扑和必须复制的其他配置数据。整个林内所有域共享一份相同的配置目录分区, 林中的每个域控制器都有一个相同的配置目录分区的副本, 对配置目录分区所做的任何更改都将复制到林中的每个域控制器。如图中的 CN=Configuration,DC=xie,DC=com 就是配置目录分区。
- 架构目录分区(Schema Directory Partition): 架构目录分区包含所有类和所有属性的条目对象, 这些条目对象定义了林中可以使用的类和属性的类型。整个林内所有域共享一份相同的架构目录分区, 林中的每个域控制器都有一个相同的架构目录分区的副本, 对架构目录分区所做的任何更改都将复制到林中的每个域控制器。因为架构目录分区规定了信息的存储方式, 因此在执行测试后, 应该在必要时通过严格控制的过程对架构目录分区进行更改, 以确保不会对林中的其他部分产生不利影响。如图中的 DC=Schema,CN=Configuration,DC=xie,DC=com 就是架构目录分区。
- 应用程序目录分区(Application Directory Partition): 从 Windows Server 2003 开始, 微软引入了应用程序目录分区, 其允许用户自定义分区来扩展目录分区。它提供了控制复制范围的能力, 并允许以更适合动态数据的方式放置副本。应用程序目录分区会被复制到林中特定的域控中, 而不是所有的域控中。如图中的 DC=DomainDnsZones,DC=xie,DC=com 和 DC=ForestDnsZones,DC=xie,DC=com 就是应用程序目录分区。

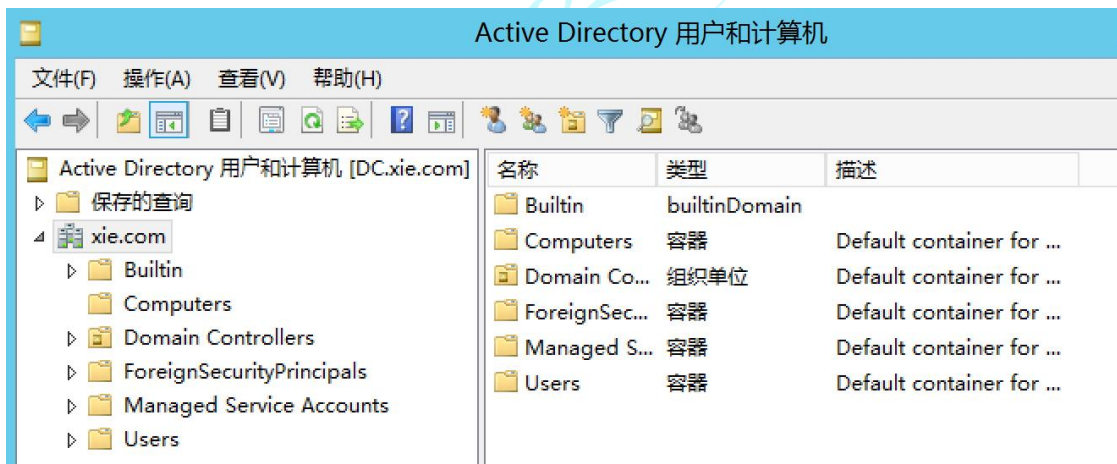
我们使用 ADEplorer 连接活动目录数据库就可以看到不同的目录分区, 如图所示, 用红色框住的分别是域目录分区、配置目录分区和架构目录分区, 而后面两个则是应用程序目录分区。



## Domain Directory Partition

每一个域各有一个域目录分区(Domain Directory Partition)，不同的域有不同的域目录分区，域目录分区包含与本地域相关联的目录对象，如用户和计算机等。我们打开 Active Directory 用户和计算机查看的默认分区就是域目录分区。

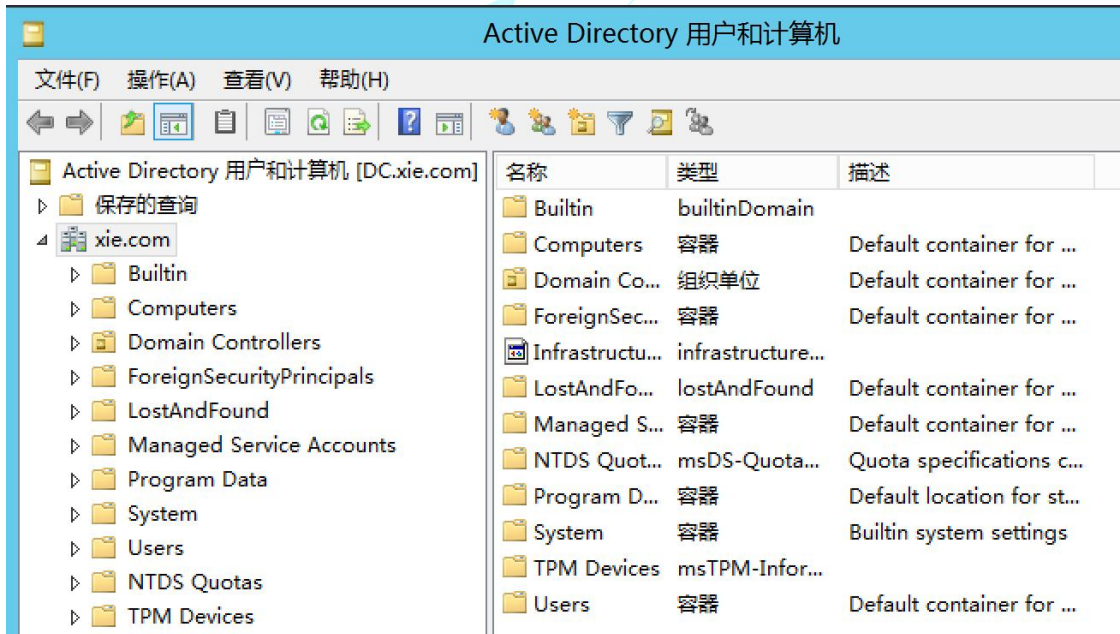
如图所示，打开“Active Directory 用户和计算机”查看到的就是域目录分区。



点击查看——>高级功能，如图所示：

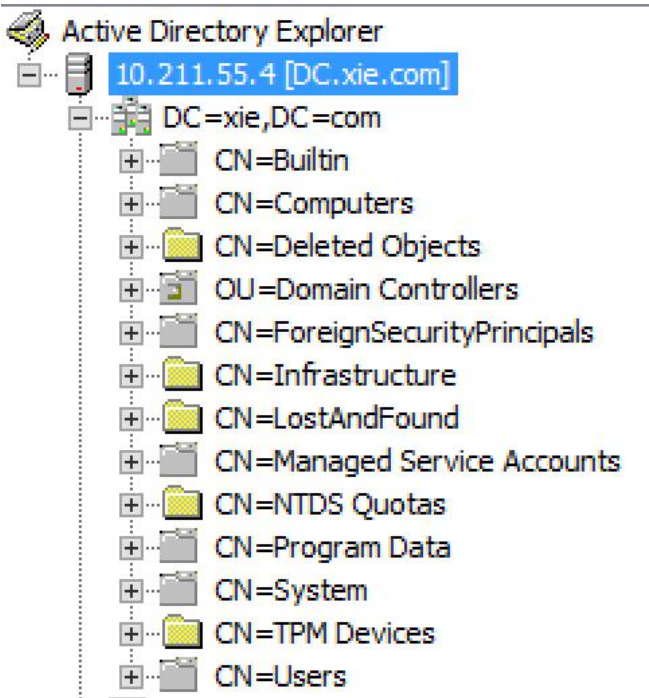


可以看到域目录分区的更多对象，如图所示：





如果使用 AD Explorer 查看域目录分区的话，默认会比“Active Directory 用户和计算机”查看的东西要多，如图 2-289 所示：



### 1. 域目录分区的顶级对象

如下表所示，是域目录分区的顶级对象以及这些顶级对象的说明：

| RDN                          | 说明                             |
|------------------------------|--------------------------------|
| CN=Builtin                   | 该容器内置了本地域组的安全组                 |
| CN=Computers                 | 机器用户的容器，包括加入域的所有机器             |
| OU=Domain Controllers        | 域控制器组织单位，包括域内所有域控              |
| CN=ForeignSecurityPrincipals | 代表域中来自森林外部域的组中的成员              |
| CN=Managed Service Accounts  | 托管服务帐户的容器                      |
| CN=System                    | 各种预配置对象的容器。包括信任对象，DNS 对象和组策略对象 |

CN=Users

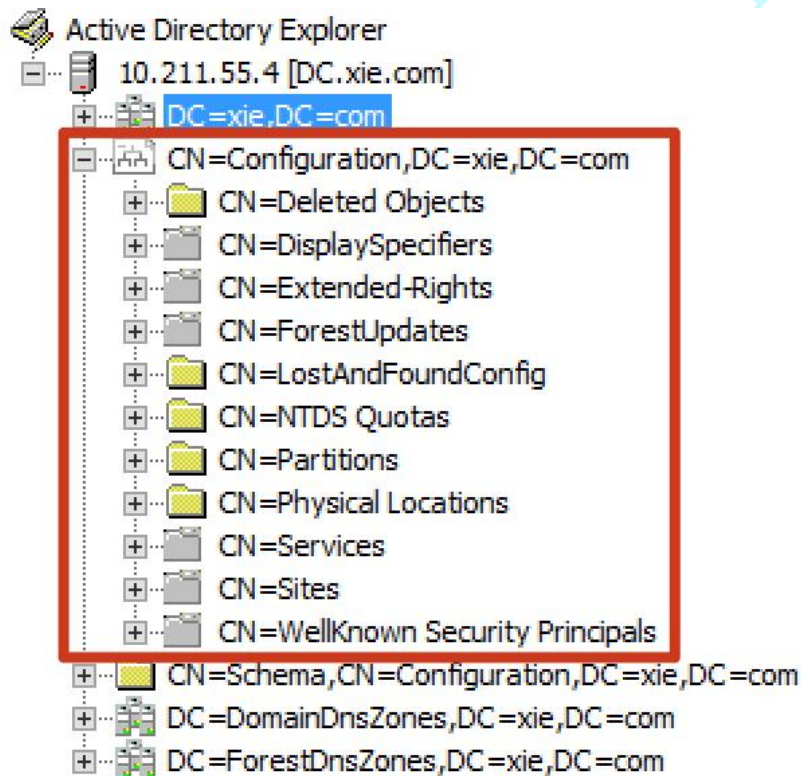
## 用户和组对象的默认容器

## Configuration Directory Partition

配置目录分区(Configuration Directory Partition)包含复制拓扑和必须复制的其他配置数据，整个林内所有域共享一份相同的配置目录分区。配置目录分区是林根域的子容器。例如，xie.com 林的配置目录分区为

CN=Configuration,DC=xie,DC=com。使用“Active Directory 用户和计算机”默认无法查看到配置目录分区。

如图所示，使用 AD Explorer 查看配置目录分区。

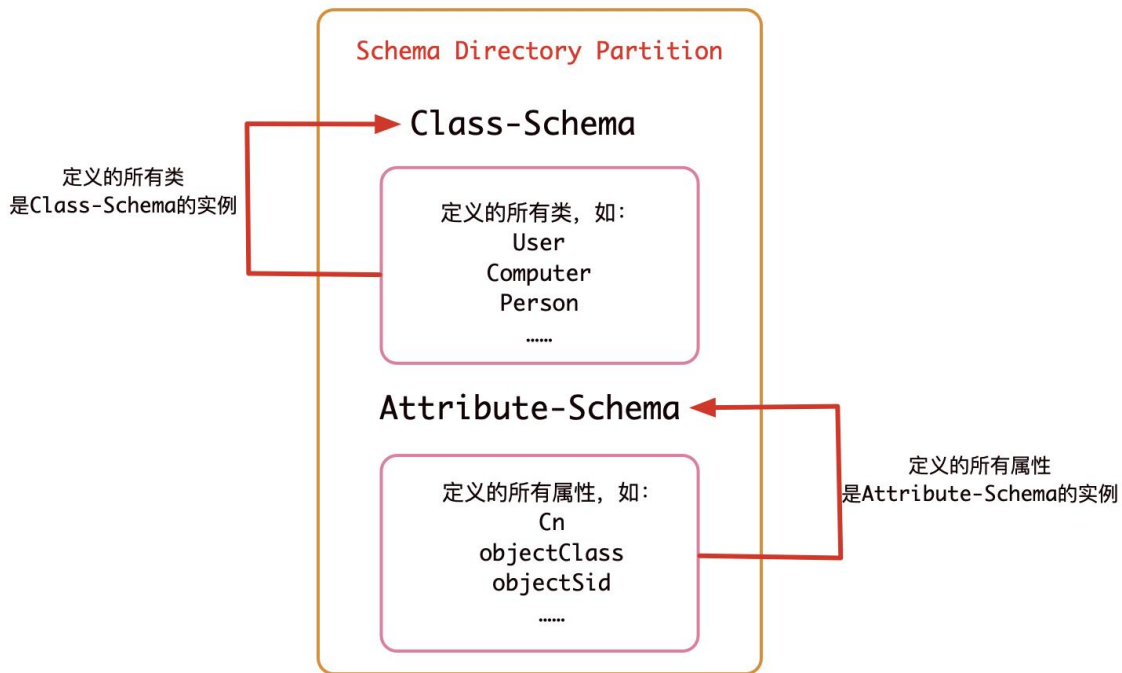


|                                  |                                                                |
|----------------------------------|----------------------------------------------------------------|
| CN=DisplaySpecifiers             | 定义了 Active Directory 管理单元的各种显示格式                               |
| CN=Extended-Rights               | 扩展权限对象的容器，我们将在域内 ACL 那篇文章里面详解                                  |
| CN=ForestUpdates                 | 包含用于表示森林状态和与域功能级别更改的对象                                         |
| CN=Partitions                    | 包含每个 Naming Context, Application Partitions 以及外部 LDAP 目录引用的对象  |
| CN=Physical Locations            | 包含位置对象，可以将其与其他对象关联 以表示该对象的位置。                                  |
| CN=Services                      | 存储有关服务的配置信息，比如文件复制服务                                           |
| CN=Sites                         | 包含所有站点拓扑和复制对象                                                  |
| CN=WellKnown Security Principals | 包含常用的外部安全性主题的对象，比如 Anonymous, Authenticated Users, Everyone 等等 |

## Schema Directory Partition

架构目录分区(Schema Directory Partition)包含所有类和所有属性的条目对象，这些条目对象定义了林中可以使用的类和属性的类型。整个林内所有域共享一份相同的架构目录分区，林中的每个域控制器都有一个相同的架构目录分区的副本，对架构目录分区所做的任何更改都将复制到林中的每个域控制器。因为架构目录分区规定了信息的存储方式，所以在执行测试后，应该在必要时通过严格控制的过程对架构目录分区进行更改，以确保不会对林的其他部分产生不利影响。与域目录分区和配置目录分区不同的是，架构目录分区不维护容器或组织单位的层次结构。相反，它是具有 Class-Schema , Attribute-Schema 和 subSchema 对象的单个容器。活动目录数据库中所有类都是 Class-Schema 的实例，而所有属性是 Attribute-Schema 的实例。也就是说，架构目录分区定义了 Active Directory 中使用的所有类和属性。

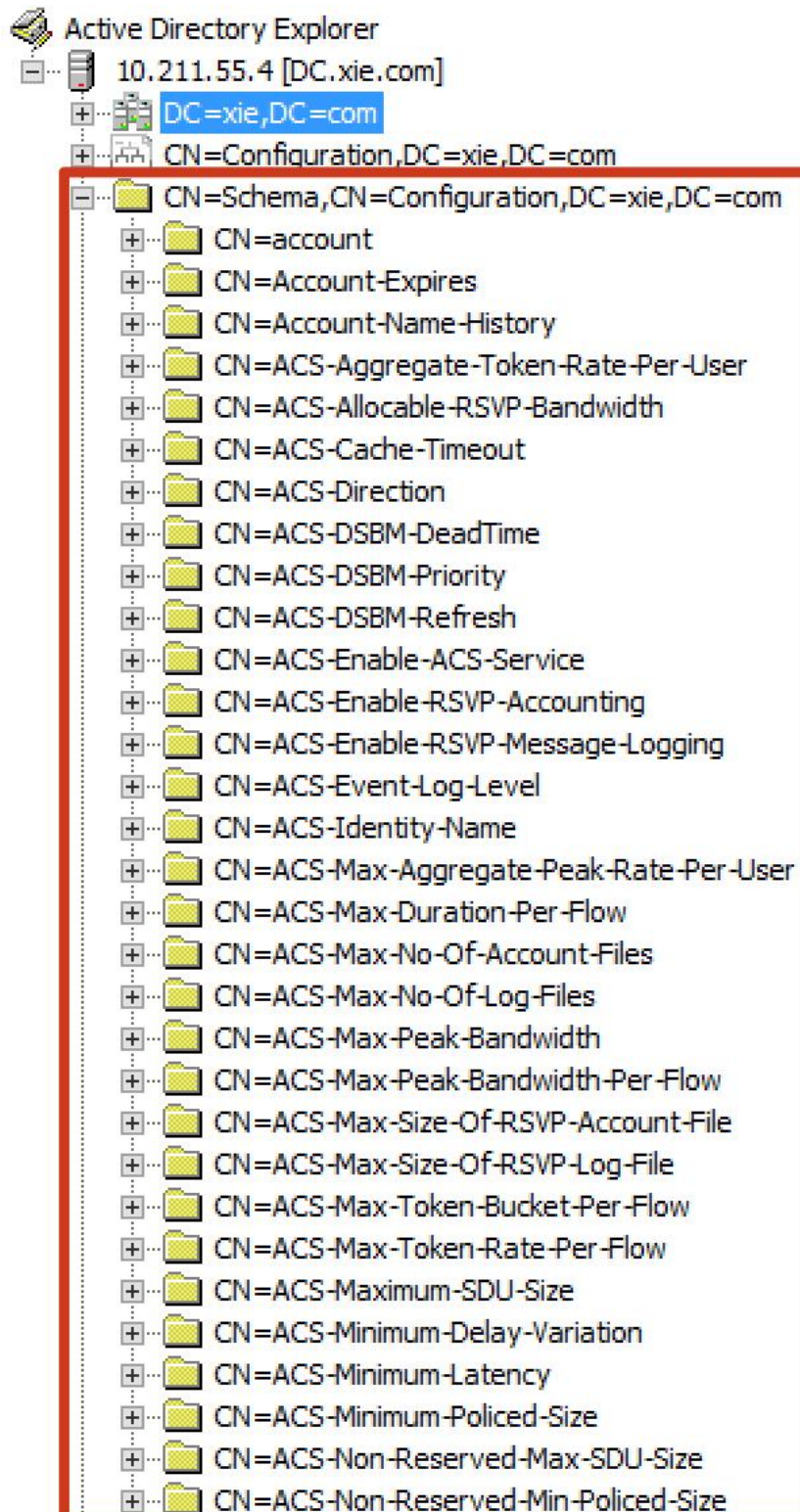
如图所示，可以看到域内定义的所有属性和类与 Class-Schema、Attribute-Schema 的关系。



## 1. 查看 Schema Directory Partition

“Active Directory 用户和计算机”默认无法查看到架构目录分区，因此如果我们要查看架构目录分区的内容，可以使用 AD Explorer。

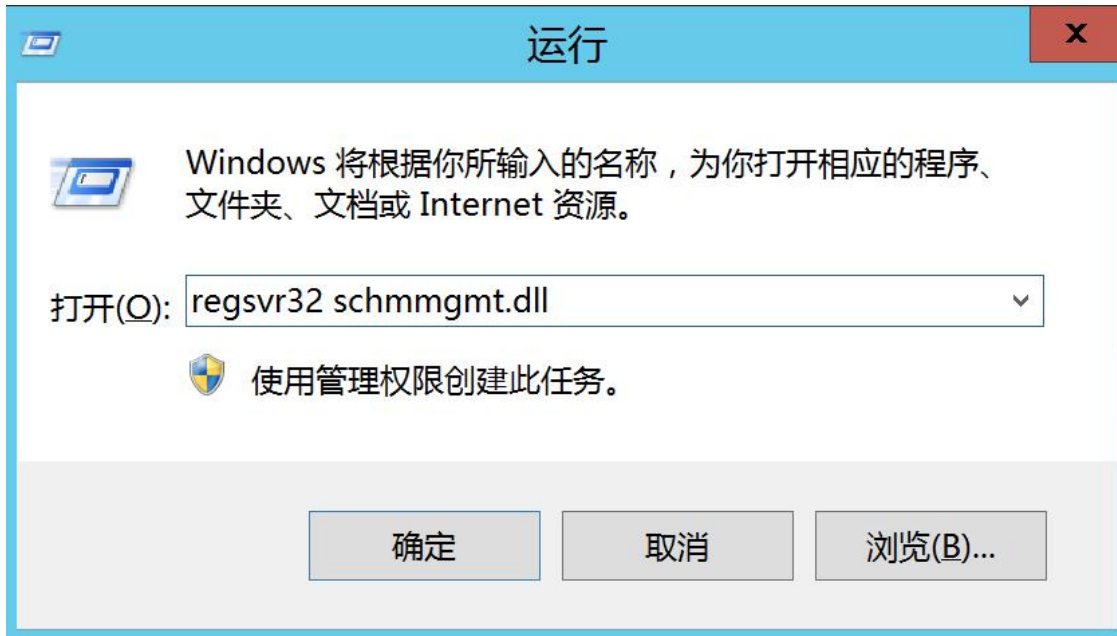
如图所示，使用 ADEplorer 连接查看架构目录分区。





也可以使用微软自带的 Active Directory Schema，但是默认没有注册该功能。可以通过运行 `regsvr32 schmmgmt.dll` 命令注册该 dll。

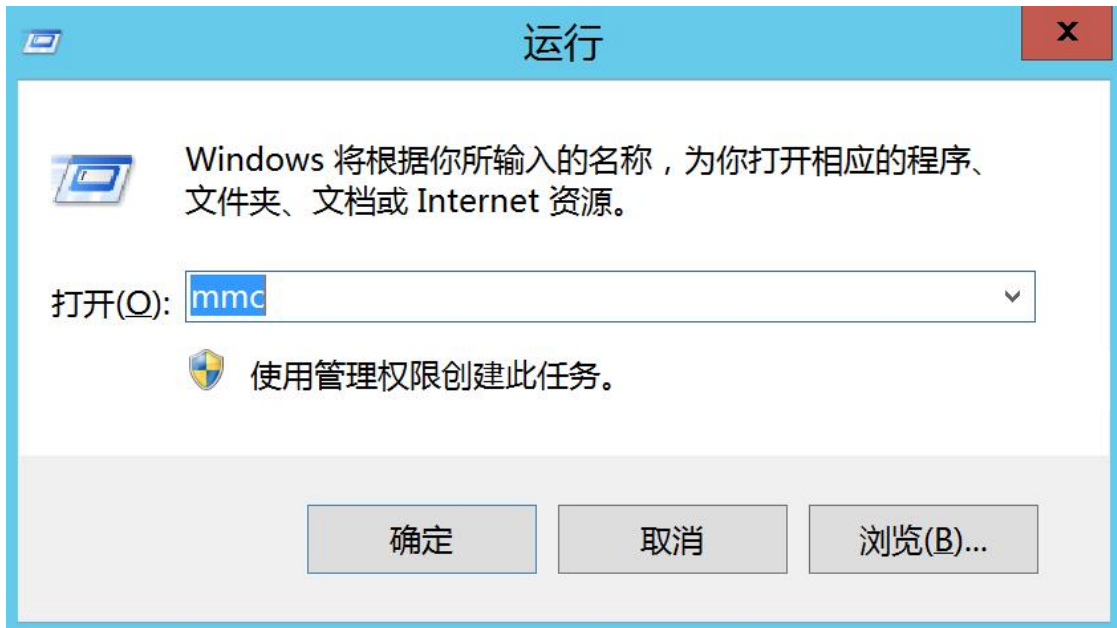
如图所示，运行 `regsvr32 schmmgmt.dll` 命令。



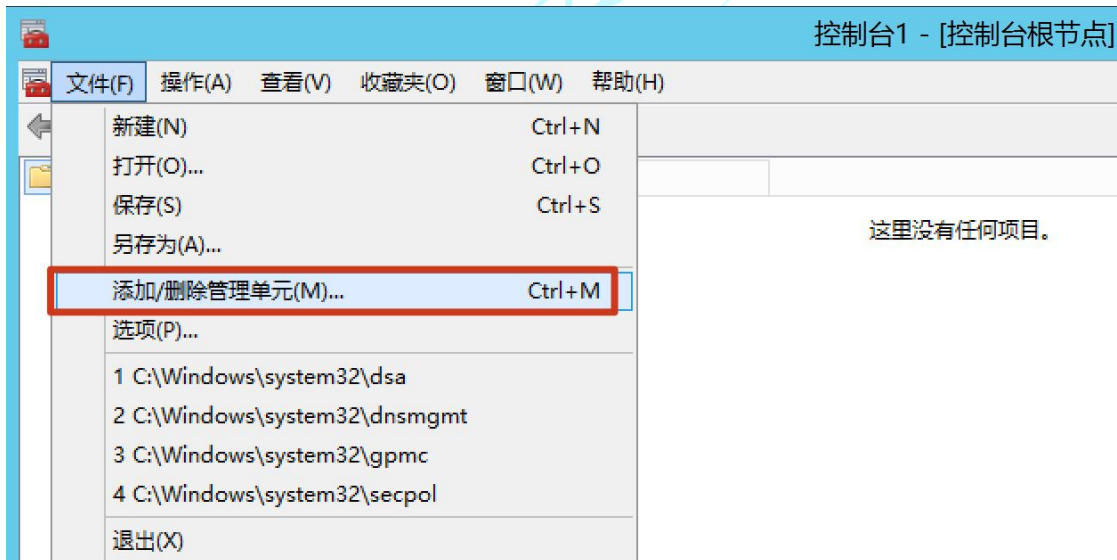
即可看到 schmmgmt.dll 注册成功，如图所示。



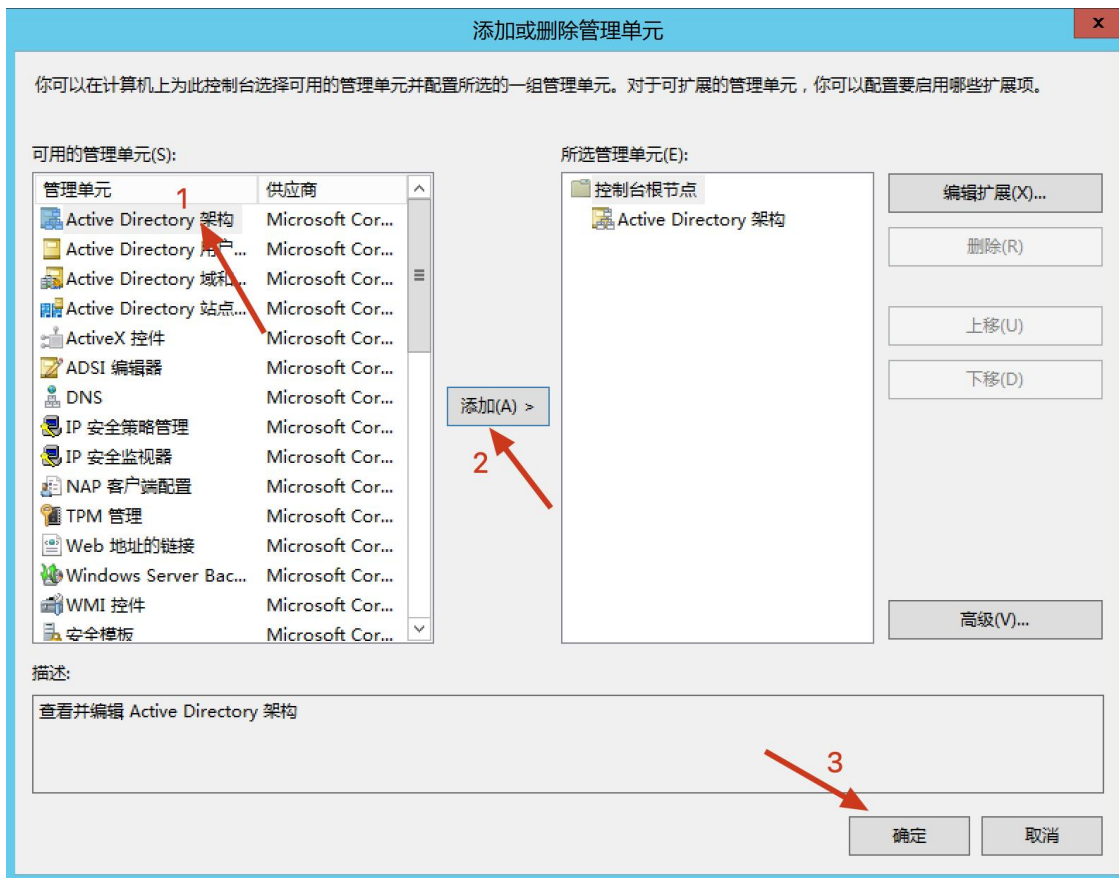
然后运行 `mmc` 打开控制台，如图所示。



在弹出的控制台，点击文件——> 添加/删除管理单元(M)，如图所示。



然后点击 Active Directory 架构——>添加——>确定，如图所示：



即可看到活动目录中定义的所有类和属性了，如图所示：

| 控制台1 - [控制台根节点\Active Directory 架构 [DC.xie.com]\类]                                                                              |       |    |                            |  |
|---------------------------------------------------------------------------------------------------------------------------------|-------|----|----------------------------|--|
| 文件(F) 操作(A) 查看(V) 收藏夹(O) 窗口(W) 帮助(H)                                                                                            |       |    |                            |  |
| <div> <div>控制台根节点</div> <div> <div>Active Directory 架构 [DC.xie.com]</div> <div> <div>类</div> <div>属性</div> </div> </div> </div> |       |    |                            |  |
| 名称                                                                                                                              | 类型    | 状态 | 描述                         |  |
| account                                                                                                                         | 结构    | 活动 | The account object cl...   |  |
| aCSPolicy                                                                                                                       | 结构    | 活动 | ACS-Policy                 |  |
| aCSResourceLimits                                                                                                               | 结构    | 活动 | ACS-Resource-Limits        |  |
| aCSSubnet                                                                                                                       | 结构    | 活动 | ACS-Subnet                 |  |
| addressBookContai...                                                                                                            | 结构    | 活动 | Address-Book-Contai...     |  |
| addressTemplate                                                                                                                 | 结构    | 活动 | Address-Template           |  |
| applicationEntity                                                                                                               | 结构    | 活动 | Application-Entity         |  |
| applicationProcess                                                                                                              | 结构    | 活动 | Application-Process        |  |
| applicationSettings                                                                                                             | 抽象    | 活动 | Application-Settings       |  |
| applicationSiteSettin...                                                                                                        | 抽象    | 活动 | Application-Site-Settin... |  |
| applicationVersion                                                                                                              | 结构    | 活动 | Stores versioning info...  |  |
| attributeSchema                                                                                                                 | 结构    | 活动 | Attribute-Schema           |  |
| bootableDevice                                                                                                                  | 辅助    | 活动 | A device with boot pa...   |  |
| builtinDomain                                                                                                                   | 结构    | 活动 | Builtin-Domain             |  |
| categoryRegistration                                                                                                            | 结构    | 活动 | Category-Registration      |  |
| certificationAuthority                                                                                                          | 类型 88 | 活动 | Certification-Authority    |  |
| classRegistration                                                                                                               | 结构    | 活动 | Class-Registration         |  |
| classSchema                                                                                                                     | 结构    | 活动 | Class-Schema               |  |
| classStore                                                                                                                      | 结构    | 活动 | Class-Store                |  |
| comConnectionPoint                                                                                                              | 结构    | 活动 | Com-Connection-Point       |  |
| computer                                                                                                                        | 结构    | 活动 | Computer                   |  |
| configuration                                                                                                                   | 结构    | 活动 | Configuration              |  |
| connectionPoint                                                                                                                 | 抽象    | 活动 | Connection-Point           |  |
| contact                                                                                                                         | 结构    | 活动 | Contact                    |  |
| container                                                                                                                       | 结构    | 活动 | Container                  |  |
| controlAccessRight                                                                                                              | 结构    | 活动 | Control-Access-Right       |  |
| country                                                                                                                         | 类型 88 | 活动 | Country                    |  |

## 2. LDAP 中的类和继承

在详细讲架构目录分区之前我们先来讲一下 LDAP 里面的类和继承，LDAP 里面的类和继承，跟开发里面的面向对象一样。

继承，也被称为派生，是指从现有对象类构建新对象类。新对象类称为其父对象的子类。所有结构对象类都直接或间接是抽象对象类 Top 的子类。在目录中表示的每个对象都属于 Top，因此每个条目都必须具有一个对象类属性。创建新类时，必须指定超类。如果不创建现有类的子类，则新类是 Top 的子类。新类可以从多个现有类继承强制属性和可选属性。但是，任何附加的类都必须由辅助类属性指定。如果将另一个属性添加到具有子类或辅助子类的类中，则在更新 Schema 缓存后，新属性将自动添加到子类中。这样说起来有点抽象，且看后文分析。

比如域内机器 CN=WIN2012R2,CN=Computers,DC=xie,DC=com 在 Active Directory 里面是一个条目对象，里面有众多属性描述该机器对象 WIN2012R2 的一些信息。

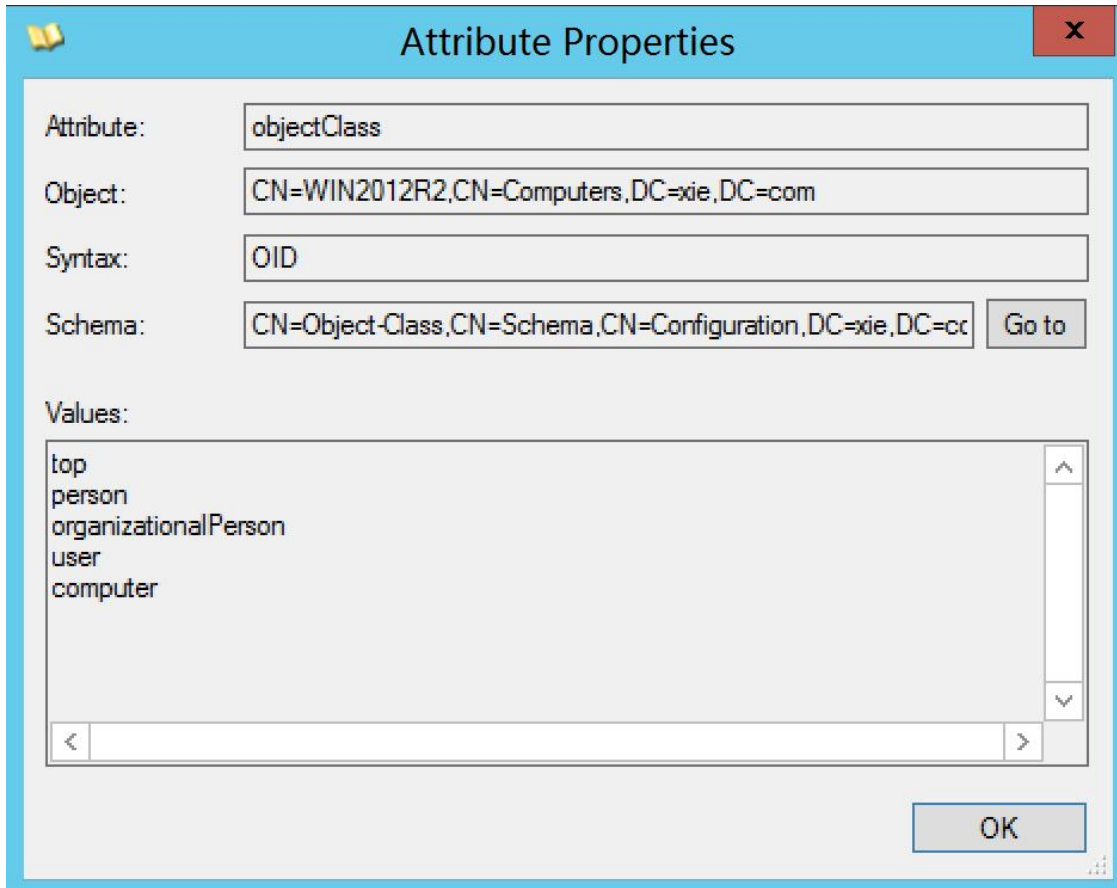
如图所示，查看机器 WIN2012R2 的属性。

| Attribute               | Syntax               | Count | Value(s)                                                                                               |
|-------------------------|----------------------|-------|--------------------------------------------------------------------------------------------------------|
| accountExpires          | Integer8             | 1     | 0x7FFFFFFFFFFFFFFF                                                                                     |
| badPasswordTime         | Integer8             | 1     | 0x0                                                                                                    |
| badPwdCount             | Integer              | 1     | 0                                                                                                      |
| cn                      | DirectoryString      | 1     | WIN2012R2                                                                                              |
| codePage                | Integer              | 1     | 0                                                                                                      |
| countryCode             | Integer              | 1     | 0                                                                                                      |
| distinguishedName       | DN                   | 1     | CN=WIN2012R2,CN=Computers,DC=xie,DC=com                                                                |
| dnsHostName             | DirectoryString      | 1     | Win2012R2.xie.com                                                                                      |
| dsCorePropagationData   | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                       |
| instanceType            | Integer              | 1     | 4                                                                                                      |
| isCriticalSystemObject  | Boolean              | 1     | FALSE                                                                                                  |
| lastLogoff              | Integer8             | 1     | 0x0                                                                                                    |
| lastLogon               | Integer8             | 1     | 2021/10/26 22:11:51                                                                                    |
| lastLogonTimestamp      | Integer8             | 1     | 2021/10/26 22:10:25                                                                                    |
| localPolicyFlags        | Integer              | 1     | 0                                                                                                      |
| logonCount              | Integer              | 1     | 9                                                                                                      |
| msDS-CreatorSID         | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-1105                                                         |
| msDS-SupportedEncryp... | Integer              | 1     | 28                                                                                                     |
| name                    | DirectoryString      | 1     | WIN2012R2                                                                                              |
| ntSecurityDescriptor    | NTSecurityDescriptor | 1     | D:(OA;;WP;5F202010-79a5-11d0-9020-00c04fc2d4cf;b967a86-0de6-11d0-a285-00aa003049e2;S-1-5-21-1313979556 |
| objectCategory          | DN                   | 1     | CN=Computer,CN=Schema,CN=Configuration,DC=xie,DC=com                                                   |
| objectClass             | OID                  | 5     | top;person;organizationalPerson;user;computer                                                          |
| objectGUID              | OctetString          | 1     | {05C32CBB-6440-4C9A-9301-499558F9B8ED}                                                                 |
| objectSid               | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-1106                                                         |
| operatingSystem         | DirectoryString      | 1     | Windows Server 2012 R2 Standard                                                                        |
| operatingSystemVersion  | DirectoryString      | 1     | 6.3 (9600)                                                                                             |
| primaryGroupID          | Integer              | 1     | 515                                                                                                    |
| pwdLastSet              | Integer8             | 1     | 2021/10/16 11:50:59                                                                                    |
| sAMAccountName          | DirectoryString      | 1     | WIN2012R2\$                                                                                            |
| sAMAccountType          | Integer              | 1     | 805306369                                                                                              |

而 WIN2012R2 条目对象有哪些属性是由他继承的类决定的。条目 CN=WIN2012R2,CN=Computers,DC=xie,DC=com 是类 Computer 的实例，在 objectClass 属性中可以看到。

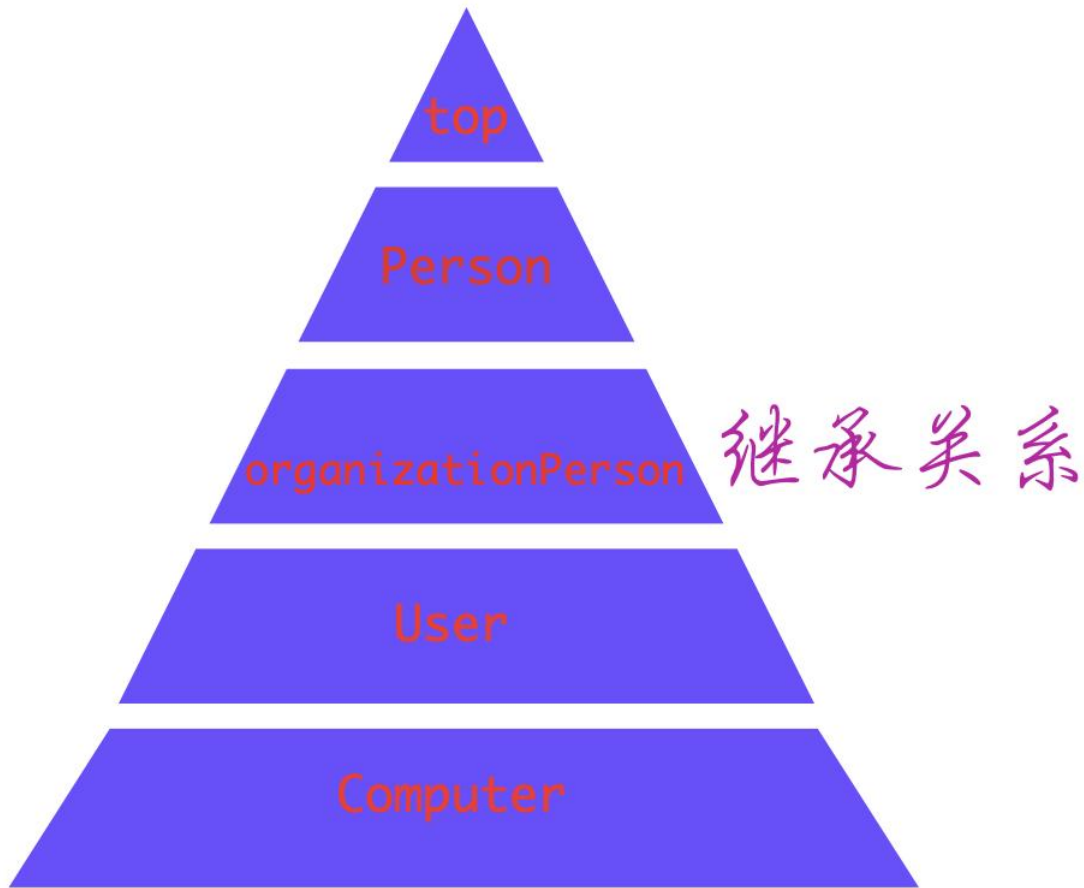
如图所示，可以看到机器 WIN2012R2 继承自 Computer。





而类是可继承的，子类继承父类的所有属性，top 类是所有类的父类。所以我们看到 objectClass 里面的值除了 computer 之外，还有 top、person、organizationPerson、user。这是因为 computer 是 user 的子类，user 是 organizationPerson 的子类，organizationPerson 是 person 的子类，而 person 是 top 的子类。因此，computer 类可以继承 top、person、organizationPerson、user 的所有属性。

如图所示，是他们之间的继承关系。



那么，computer 类默认到底继承父类的哪些属性呢？这跟父类对象的以下几个属性相关：

- systemMustContain 和 MustContain 属性里的值是必选继承的属性
- systemMayContain 和 MayContain 属性里的值是可选继承的属性

我们就以 computer 类来看。该类继承于 user——>organizationalPerson——>person——>top。

那我们一个一个来看看。

如图查看 top 类的 systemMustContain 属性，可以看到 top 类的强制继承属性如下：

- objectClass
- objectCategory
- nTSecurityDescriptor
- instanceType

| Attribute                 | Syntax               | Count | Value(s)                                                                                                                                        |
|---------------------------|----------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | Top                                                                                                                                             |
| adminDisplayName          | DirectoryString      | 1     | Top                                                                                                                                             |
| cn                        | DirectoryString      | 1     | Top                                                                                                                                             |
| defaultHidingValue        | Boolean              | 1     | TRUE                                                                                                                                            |
| defaultObjectCategory     | DN                   | 1     | CN=Top,CN=Schema,CN=Configuration,DC=xie,DC=com                                                                                                 |
| defaultSecurityDescriptor | DirectoryString      | 1     | D:(A;;RPWPCCDCLCLORCWOWSDOTS;;;DA)(A;;RPWPCCDCLCLORCWOWSDOTS;;;SY)(A;;RPLCLORC;;;AU)                                                            |
| distinguishedName         | DN                   | 1     | CN=Top,CN=Schema,CN=Configuration,DC=xie,DC=com                                                                                                 |
| dsCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                                                                |
| governorID                | OID                  | 1     | 2.5.6.0                                                                                                                                         |
| instanceType              | Integer              | 1     | 4                                                                                                                                               |
| LDAPDisplayName           | DirectoryString      | 1     | top                                                                                                                                             |
| mayContain                | OID                  | 4     | msSFU30PosixMemberOf;msDFSR-ComputerReferenceBL;msDFSR-MemberReferenceBL;msDS-ObjectReferenceBL                                                 |
| name                      | DirectoryString      | 1     | Top                                                                                                                                             |
| nTSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIID;LCPLORC;;;AU)(A;CIID;CCLCSWRPWPLOCRRCWDWO;;;SA)(A;CIID;CCDCLCSWRPWPDTLOCRSDRCWDWO;;;SY)                                             |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                                                        |
| objectClass               | OID                  | 2     | top;classSchema                                                                                                                                 |
| objectClassCategory       | Enumeration          | 1     | 2                                                                                                                                               |
| objectGUID                | OctetString          | 1     | {4C745911-D477-4C51-A9AC-9EA28AFD5765}                                                                                                          |
| rDNAHID                   | OID                  | 1     | cn                                                                                                                                              |
| schemaIDGUID              | OctetString          | 1     | {BF967AB7-0DE6-11D0-A285-00AA003049E2}                                                                                                          |
| showInAdvancedViewO...    | Boolean              | 1     | TRUE                                                                                                                                            |
| subClassOf                | OID                  | 1     | top                                                                                                                                             |
| systemFlags               | Integer              | 1     | 16                                                                                                                                              |
| systemMayContain          | OID                  | 114   | msDS-RepValueMetaDataType;msDS-parentsDistname;msds-memberTransitive;msds-memberOfTransitive;msDS-TDOEGressBL;msDS-TDOIgressBL;msDS-ValueTyp... |
| systemMustContain         | OID                  | 4     | objectClass;objectCategory;nTSecurityDescriptor;instanceType                                                                                    |
| systemOnly                | Boolean              | 1     | TRUE                                                                                                                                            |
| systemPossSuperiors       | OID                  | 1     | lostAndFound                                                                                                                                    |
| usnChanged                | Integer8             | 1     | 0x751                                                                                                                                           |
| usnCreated                | Integer8             | 1     | 0x751                                                                                                                                           |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                                                               |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                                                               |

如图查看 person 类的 systemMustContain 属性，可以看到 person 类的强制继承属性为 cn

| Attribute                 | Syntax               | Count | Value(s)                                                                                            |
|---------------------------|----------------------|-------|-----------------------------------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | Person                                                                                              |
| adminDisplayName          | DirectoryString      | 1     | Person                                                                                              |
| cn                        | DirectoryString      | 1     | Person                                                                                              |
| defaultHidingValue        | Boolean              | 1     | TRUE                                                                                                |
| defaultObjectCategory     | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                                                  |
| defaultSecurityDescriptor | DirectoryString      | 1     | D:(A;;RPWPCCDCLCLORCWOWSDSDTSW;;;DA)(A;;RPWPCCDCLCLORCWOWSDSDTSW;;;SY)(A;;RPLCLORC;;;AU)            |
| distinguishedName         | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                                                  |
| dSCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                    |
| governsID                 | OID                  | 1     | 2.5.6.6                                                                                             |
| instanceType              | Integer              | 1     | 4                                                                                                   |
| DAPDisplayName            | DirectoryString      | 1     | person                                                                                              |
| mayContain                | OID                  | 1     | attributeCertificateAttribute                                                                       |
| name                      | DirectoryString      | 1     | Person                                                                                              |
| nTSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIID;LCRPLORC;;;AU)(A;CIID;CCLCSWRPWPLOCRRCWDWO;;;SA)(A;CIID;CCDCLCSWRPWPDTLOCRSDRCWDWO;;;S) |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                            |
| objectClass               | OID                  | 2     | top;classSchema                                                                                     |
| objectClassCategory       | Enumeration          | 1     | 0                                                                                                   |
| objectGUID                | OctetString          | 1     | {8019BCF6-0A33-48A6-B2F6-092F628F572F}                                                              |
| rDNAtID                   | OID                  | 1     | cn                                                                                                  |
| schemalDGUID              | OctetString          | 1     | {BF967AA7-0DE6-11D0-A285-00AA003049E2}                                                              |
| showInAdvancedViewO...    | Boolean              | 1     | TRUE                                                                                                |
| subClassOf                | OID                  | 1     | top                                                                                                 |
| systemFlags               | Integer              | 1     | 16                                                                                                  |
| systemMayContain          | OID                  | 5     | userPassword;telephoneNumber;sn;serialNumber;seeAlso                                                |
| systemMustContain         | OID                  | 1     | cn                                                                                                  |
| systemOnly                | Boolean              | 1     | FALSE                                                                                               |
| systemPossSuperiors       | OID                  | 2     | organizationalUnit;container                                                                        |
| uSNChanged                | Integer8             | 1     | 0x6FA                                                                                               |
| uSNCreated                | Integer8             | 1     | 0x6FA                                                                                               |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                   |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                   |

如图查看 organizationalPerson 类的 systemMustContain 属性，可以看到 organizationalPerson 类没有强制属性。

| Attribute                 | Syntax               | Count | Value(s)                                                                                                               |
|---------------------------|----------------------|-------|------------------------------------------------------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | Organization                                                                                                           |
| adminDisplayName          | DirectoryString      | 1     | Organization                                                                                                           |
| cn                        | DirectoryString      | 1     | Organization                                                                                                           |
| defaultHidingValue        | Boolean              | 1     | FALSE                                                                                                                  |
| defaultObjectCategory     | DN                   | 1     | CN=Organization,CN=Schema,CN=Configuration,DC=xie,DC=com                                                               |
| defaultSecurityDescriptor | DirectoryString      | 1     | D:(A;;RPWPCCDCLCLORCWOWSDSDTSW;;;DA)(A;;RPWPCCDCLCLORCWOWSDSDTSW;;;SY)(A;;RPLCLORC;;;AU)                               |
| distinguishedName         | DN                   | 1     | CN=Organization,CN=Schema,CN=Configuration,DC=xie,DC=com                                                               |
| dSCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                                       |
| governsID                 | OID                  | 1     | 2.5.6.4                                                                                                                |
| instanceType              | Integer              | 1     | 4                                                                                                                      |
| DAPDisplayName            | DirectoryString      | 1     | organization                                                                                                           |
| name                      | DirectoryString      | 1     | Organization                                                                                                           |
| nTSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIID;LCRPLORC;;;AU)(A;CIID;CCLCSWRPWPLOCRRCWDWO;;;SA)(A;CIID;CCDCLCSWRPWPDTLOCRSDRCWDWO;;;S)                    |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                               |
| objectClass               | OID                  | 2     | top;classSchema                                                                                                        |
| objectClassCategory       | Enumeration          | 1     | 1                                                                                                                      |
| objectGUID                | OctetString          | 1     | {2B048D27-76D3-4F55-AD30-497C30421059}                                                                                 |
| rDNAtID                   | OID                  | 1     | o                                                                                                                      |
| schemalDGUID              | OctetString          | 1     | {BF967AA3-0DE6-11D0-A285-00AA003049E2}                                                                                 |
| showInAdvancedViewO...    | Boolean              | 1     | TRUE                                                                                                                   |
| subClassOf                | OID                  | 1     | top                                                                                                                    |
| systemFlags               | Integer              | 1     | 16                                                                                                                     |
| systemMayContain          | OID                  | 20    | x121Address;userPassword;telexNumber;teletexTerminalIdentifier;telephoneNumber;street;st;seeAlso;searchGuide;registeri |
| systemMustContain         | OID                  | 1     | o                                                                                                                      |
| systemOnly                | Boolean              | 1     | FALSE                                                                                                                  |
| systemPossSuperiors       | OID                  | 3     | locality;country;domainDNS                                                                                             |
| uSNChanged                | Integer8             | 1     | 0x6F2                                                                                                                  |
| uSNCreated                | Integer8             | 1     | 0x6F2                                                                                                                  |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                                      |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                                      |

如图查看 user 类的 systemMustContain 属性，可以看到 user 类也没有强制继承属性。

| Attribute                 | Syntax               | Count | Value(s)                                                                                          |
|---------------------------|----------------------|-------|---------------------------------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | User                                                                                              |
| adminDisplayName          | DirectoryString      | 1     | User                                                                                              |
| auxiliaryClass            | OID                  | 2     | shadowAccount;posixAccount                                                                        |
| cn                        | DirectoryString      | 1     | User                                                                                              |
| defaultHidingValue        | Boolean              | 1     | FALSE                                                                                             |
| defaultObjectCategory     | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                                                |
| defaultSecurityDescriptor | DirectoryString      | 1     | D:(A;;RPWPCRCCLCLORCWOWSDOTS;;;DA)(A;;RPWPCRCCLCLORCWOWSDOTS;;;SY)(A;;RPW                         |
| distinguishedName         | DN                   | 1     | CN=User,CN=Schema,CN=Configuration,DC=xie,DC=com                                                  |
| dSCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                  |
| governsID                 | OID                  | 1     | 1.2.840.113556.1.5.9                                                                              |
| instanceType              | Integer              | 1     | 4                                                                                                 |
| IDAPDisplayName           | DirectoryString      | 1     | user                                                                                              |
| mayContain                | OID                  | 21    | msDS-SourceObjectDN;msSFU30NisDomain;msSFU30Name;x500uniqueIdentifier;userSMIMECertificate;userF  |
| name                      | DirectoryString      | 1     | User                                                                                              |
| nTSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIID;LCRPLORC;;;AU)(A;CIID;CCLCSWRPLOCRRRCWDWO;;;SA)(A;CIID;CCDCLCSWRPDTLOC                |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                          |
| objectClass               | OID                  | 2     | top;classSchema                                                                                   |
| objectClassCategory       | Enumeration          | 1     | 1                                                                                                 |
| objectGUID                | OctetString          | 1     | {E1ECD7D4-3A94-4848-B95B-2469917D6D5F}                                                            |
| rDNAttID                  | OID                  | 1     | cn                                                                                                |
| schemaIDGUID              | OctetString          | 1     | {BF967ABA-0DE6-11D0-A285-00AA003049E2}                                                            |
| showInAdvancedViewO...    | Boolean              | 1     | TRUE                                                                                              |
| subClassOf                | OID                  | 1     | organizationalPerson                                                                              |
| systemAuxiliaryClass      | OID                  | 3     | msDS-CloudExtensions;securityPrincipal;mailRecipient                                              |
| systemFlags               | Integer              | 1     | 16                                                                                                |
| systemMayContain          | OID                  | 133   | msDS-AuthNPolicySiloMembersBL;msDS-AssignedAuthNPolicySilo;msDS-AssignedAuthNPolicy;msDS-SyncServ |
| systemOnly                | Boolean              | 1     | FALSE                                                                                             |
| systemPossSuperiors       | OID                  | 3     | builtinDomain;organizationalUnit;domainDNS                                                        |
| uSNCchanged               | Integer8             | 1     | 0x756                                                                                             |
| uSNCreated                | Integer8             | 1     | 0x756                                                                                             |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                 |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                 |

因此通过继承，computer 类的强制继承属性有如下：

- objectClass
- objectCategory
- nTSecurityDescriptor
- instanceType
- cn

其他属性的话都是可选继承的。

我们通过 Active DirectorySchema 查看，就会看到强制继承的属性和可选继承的属性，以及源类是哪个类。



如图所示，可以看到 computer 类默认强制继承的有如下属性。

| 名称                   | 类型 | 系统 | 描述                            | 源类                |
|----------------------|----|----|-------------------------------|-------------------|
| sAMAccountName       | 强制 | 是  | SAM-Account-Name              | securityPrincipal |
| objectSid            | 强制 | 是  | Object-Sid                    | securityPrincipal |
| cn                   | 强制 | 是  | Common-Name                   | mailRecipient     |
| cn                   | 强制 | 是  | Common-Name                   | person            |
| objectClass          | 强制 | 是  | Object-Class                  | top               |
| objectCategory       | 强制 | 是  | Object-Category               | top               |
| nTSecurityDescriptor | 强制 | 是  | NT-Security-Descriptor        | top               |
| instanceType         | 强制 | 是  | Instance-Type                 | top               |
| msImaging-HashAl...  | 可选 | 是  | Contains the name of th...    | computer          |
| msImaging-Thumb...   | 可选 | 是  | Contains a hash of the s...   | computer          |
| msDS-GenerationId    | 可选 | 是  | For virtual machine sna...    | computer          |
| msTPM-TpmInform...   | 可选 | 是  | This attribute links a Co...  | computer          |
| msTSSecondaryDes...  | 可选 | 是  | This attribute represent...   | computer          |
| msTSPrimaryDesko...  | 可选 | 是  | This attribute represent...   | computer          |
| msTSEndpointPlug...  | 可选 | 是  | This attribute represent...   | computer          |
| msTSEndpointType     | 可选 | 是  | This attribute defines if ... | computer          |
| msTSEndpointData     | 可选 | 是  | This attribute represent...   | computer          |
| msDS-HostServiceA... | 可选 | 是  | Service Accounts config...    | computer          |
| msDS-IsUserCacha...  | 可选 | 是  | For a Read-Only Direct...     | computer          |
| msTSProperty02       | 可选 | 是  | Placeholder Terminal Se...    | computer          |

为什么会比我们上面查询的强制继承的属性还要多了来自于 securityPrincipal 源类的 sAMAccountName 属性和 objectSid 属性呢以及 mailRecipient 源类的 cn 属性呢？

如图所示，可以看到 user 类有个 systemAuxiliaryClass 属性(也就是辅助类)，而该辅助类属性的值也是需要继承的。因此 computer 类还需要继承来自辅助类 msDS-CloudExtensions、securityPrincipal 和 mailRecipient 的强制继承属性。

Path: CN=User,CN=Schema,CN=Configuration,DC=xie,DC=com,10.211.55.4 [DC.xie.com]

| Attribute                 | Syntax               | Count | Value(s)                                                                                                  |
|---------------------------|----------------------|-------|-----------------------------------------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | User                                                                                                      |
| adminDisplayName          | DirectoryString      | 1     | User                                                                                                      |
| auxiliaryClass            | OID                  | 2     | shadowAccount;posixAccount                                                                                |
| cn                        | DirectoryString      | 1     | User                                                                                                      |
| defaultHidingValue        | Boolean              | 1     | FALSE                                                                                                     |
| defaultObjectCategory     | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                                                        |
| defaultSecurityDescriptor | DirectoryString      | 1     | D: (A;;RPWPCRCDCDCLORCOWOWSDTSW;;;DA)(A;;RPWPCRCDCDCLORCOWOWSDTSW;;;DA)(A;;RPWPCRCDCDCLORCOWOWSDTSW;;;DA) |
| distinguishedName         | DN                   | 1     | CN=User,CN=Schema,CN=Configuration,DC=xie,DC=com                                                          |
| dsCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                          |
| governorID                | OID                  | 1     | 1.2.840.113556.1.5.9                                                                                      |
| instanceType              | Integer              | 1     | 4                                                                                                         |
| LDAPDisplayName           | DirectoryString      | 1     | user                                                                                                      |
| mayContain                | OID                  | 21    | msDS-SourceObjectDN;msSFU30NsDomain;msSFU30Name;x500uniqueIdent                                           |
| name                      | DirectoryString      | 1     | User                                                                                                      |
| ntSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIID;LCRPLORC;;;AU)(A;CIID;CCLCSWRPWPLOCRRCDWOWO;;;SA)(                                            |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                  |
| objectClass               | OID                  | 2     | top;classSchema                                                                                           |
| objectClassCategory       | Enumeration          | 1     | 1                                                                                                         |
| objectGUID                | OctetString          | 1     | {E1ECD7D4-3A94-4848-B95B-2469917D6D5F}                                                                    |
| rDNAttID                  | OID                  | 1     | cn                                                                                                        |
| schemaIDGUID              | OctetString          | 1     | {BF967ABA-0DE6-11D0-A285-00AA003049E2}                                                                    |
| showInAdvancedViewOnly    | Boolean              | 1     | TRUE                                                                                                      |
| subClassOf                | OID                  | 1     | organizationalPerson                                                                                      |
| systemAuxiliaryClass      | OID                  | 3     | msDS-CloudExtensions;securityPrincipal;mailRecipient                                                      |
| systemFlags               | Integer              | 1     | 16                                                                                                        |
| systemMayContain          | OID                  | 133   | msDS-AuthnPolicySiloMembersBL;msDS-AssignedAuthnPolicySilo;msDS-Assig                                     |
| systemOnly                | Boolean              | 1     | FALSE                                                                                                     |
| systemPossSuperiors       | OID                  | 3     | builtinDomain;organizationalUnit;domainDNS                                                                |
| uSNCreated                | Integer8             | 1     | 0x756                                                                                                     |
| uSNCreated                | Integer8             | 1     | 0x756                                                                                                     |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                         |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                         |

查询 msDS-CloudExtensions、securityPrincipal 和 mailRecipient 三个辅助类的强制继承属性。

如图，查看 msDS-CloudExtensions 类的 systemMustContain 属性，可以看到 msDS-CloudExtensions 类没有强制继承属性。

| Attribute                 | Syntax               | Count | Value(s)                                                                           |
|---------------------------|----------------------|-------|------------------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | A collection of attributes used to house arbitrary cloud-relevant strings.         |
| adminDisplayName          | DirectoryString      | 1     | ms-DS-Cloud-Extensions                                                             |
| cn                        | DirectoryString      | 1     | ms-DS-Cloud-Extensions                                                             |
| defaultHidingValue        | Boolean              | 1     | TRUE                                                                               |
| defaultObjectCategory     | DN                   | 1     | CN=ms-DS-Cloud-Extensions,CN=Schema,CN=Configuration,DC=xie,DC=com                 |
| defaultSecurityDescriptor | DirectoryString      | 1     | D:(A;;RPWPCCDCLCLORCWOWDSDDTSW;;;DA)(A;;RPWPCCDCLCLORCWOWDSDE                      |
| distinguishedName         | DN                   | 1     | CN=ms-DS-Cloud-Extensions,CN=Schema,CN=Configuration,DC=xie,DC=com                 |
| dSCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                   |
| governsID                 | OID                  | 1     | 1.2.840.113556.1.5.283                                                             |
| instanceType              | Integer              | 1     | 4                                                                                  |
| IDAPDisplayName           | DirectoryString      | 1     | msDS-CloudExtensions                                                               |
| mayContain                | OID                  | 20    | msDS-cloudExtensionAttribute20;msDS-cloudExtensionAttribute19;msDS-cloudExtensionA |
| name                      | DirectoryString      | 1     | ms-DS-Cloud-Extensions                                                             |
| nTSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIIID;LCRPLORC;;;AU)(A;CIIID;CCLCSWRPWPLOCRRRCWDWO;;;SA)(A;CIIID;CCDC       |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                           |
| objectClass               | OID                  | 2     | top;classSchema                                                                    |
| objectClassCategory       | Enumeration          | 1     | 3                                                                                  |
| objectGUID                | OctetString          | 1     | {3C160244-D984-4E04-98E4-0DDED59F587C}                                             |
| rDNAttID                  | OID                  | 1     | cn                                                                                 |
| schemaIDGUID              | OctetString          | 1     | {641E87A4-8326-4771-8A2D-C706DF35E35A}                                             |
| showInAdvancedViewOnly    | Boolean              | 1     | TRUE                                                                               |
| subClassOf                | OID                  | 1     | top                                                                                |
| systemFlags               | Integer              | 1     | 16                                                                                 |
| systemOnly                | Boolean              | 1     | FALSE                                                                              |
| uSNChanged                | Integer8             | 1     | 0x7B9                                                                              |
| uSNCreated                | Integer8             | 1     | 0x7B9                                                                              |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                  |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                  |

如图，查看 securityPrincipal 类的 systemMustContain 属性，可以看到 securityPrincipal 类的强制继承属性为 sAMAccountName 和 objectSid。

| Attribute              | Syntax               | Count | Value(s)                                                                       |
|------------------------|----------------------|-------|--------------------------------------------------------------------------------|
| adminDescription       | DirectoryString      | 1     | Security-Principal                                                             |
| adminDisplayName       | DirectoryString      | 1     | Security-Principal                                                             |
| cn                     | DirectoryString      | 1     | Security-Principal                                                             |
| defaultHidingValue     | Boolean              | 1     | TRUE                                                                           |
| defaultObjectCategory  | DN                   | 1     | CN=Security-Principal,CN=Schema,CN=Configuration,DC=xie,DC=com                 |
| distinguishedName      | DN                   | 1     | CN=Security-Principal,CN=Schema,CN=Configuration,DC=xie,DC=com                 |
| dsCorePropagationData  | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                               |
| governsID              | OID                  | 1     | 1.2.840.113556.1.5.6                                                           |
| instanceType           | Integer              | 1     | 4                                                                              |
| IDAPDisplayName        | DirectoryString      | 1     | securityPrincipal                                                              |
| name                   | DirectoryString      | 1     | Security-Principal                                                             |
| ntSecurityDescriptor   | NTSecurityDescriptor | 1     | D:AI(A;CIID;LCRPLORC;;;AU)(A;CIID;CCLCSWRPWPLOCRRRCWDWO;;;SA)(A;CI             |
| objectCategory         | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                       |
| objectClass            | OID                  | 2     | top;classSchema                                                                |
| objectClassCategory    | Enumeration          | 1     | 3                                                                              |
| objectGUID             | OctetString          | 1     | {E6D751FE-E00C-40DA-9098-AC06C7EFE2F7}                                         |
| rDNAttID               | OID                  | 1     | cn                                                                             |
| schemaIDGUID           | OctetString          | 1     | {BF967AB0-0DE6-11D0-A285-00AA003049E2}                                         |
| showInAdvancedViewOnly | Boolean              | 1     | TRUE                                                                           |
| subClassOf             | OID                  | 1     | top                                                                            |
| systemFlags            | Integer              | 1     | 16                                                                             |
| systemMayContain       | OID                  | 12    | supplementalCredentials;sIDHistory;securityIdentifier;sAMAccountType;rid;token |
| systemMustContain      | OID                  | 2     | sAMAccountName;objectSid                                                       |
| systemOnly             | Boolean              | 1     | FALSE                                                                          |
| uSNChanged             | Integer8             | 1     | 0x732                                                                          |
| uSNCreated             | Integer8             | 1     | 0x732                                                                          |
| whenChanged            | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                              |
| whenCreated            | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                              |

如图，查看 mailRecipient 类的 systemMustContain 属性，可以看到 mailRecipient 类的强制继承属性为 cn。

| Attribute                 | Syntax               | Count | Value(s)                                                                    |
|---------------------------|----------------------|-------|-----------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | Mail-Recipient                                                              |
| adminDisplayName          | DirectoryString      | 1     | Mail-Recipient                                                              |
| cn                        | DirectoryString      | 1     | Mail-Recipient                                                              |
| defaultHidingValue        | Boolean              | 1     | TRUE                                                                        |
| defaultObjectCategory     | DN                   | 1     | CN=Mail-Recipient,CN=Schema,CN=Configuration,DC=xie,DC=com                  |
| defaultSecurityDescriptor | DirectoryString      | 1     | D:(A;;RPWPCRCCDCLORCOWDSDTSW;;;DA)(A;;RPWPCRCCDCLORCOW                      |
| distinguishedName         | DN                   | 1     | CN=Mail-Recipient,CN=Schema,CN=Configuration,DC=xie,DC=com                  |
| dsCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                            |
| governsID                 | OID                  | 1     | 1.2.840.113556.1.3.46                                                       |
| instanceType              | Integer              | 1     | 4                                                                           |
| IDAPDisplayName           | DirectoryString      | 1     | mailRecipient                                                               |
| mayContain                | OID                  | 9     | msDS-GeoCoordinatesLongitude;msDS-GeoCoordinatesLatitude;msDS-GeoCoordinat  |
| name                      | DirectoryString      | 1     | Mail-Recipient                                                              |
| nTSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIID;LCRPLORC;;;AU)(A;CIID;CCLCSWRPWPLOCRRCWDWO;;;SA)(A;CIID;        |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                    |
| objectClass               | OID                  | 2     | top;classSchema                                                             |
| objectClassCategory       | Enumeration          | 1     | 3                                                                           |
| objectGUID                | OctetString          | 1     | {0462DE0C-61EB-400E-985E-36DF1BF21E7A}                                      |
| rDNAttID                  | OID                  | 1     | cn                                                                          |
| schemaIDGUID              | OctetString          | 1     | {BF967AA1-0DE6-11D0-A285-00AA003049E2}                                      |
| showInAdvancedViewOnly    | Boolean              | 1     | TRUE                                                                        |
| subClassOf                | OID                  | 1     | top                                                                         |
| systemFlags               | Integer              | 1     | 16                                                                          |
| systemMayContain          | OID                  | 8     | userCertificate;userCert;textEncodedORAddress;telephoneNumber;showInAddress |
| systemMustContain         | OID                  | 1     | cn                                                                          |
| systemOnly                | Boolean              | 1     | FALSE                                                                       |
| systemPossSuperiors       | OID                  | 1     | container                                                                   |
| uSNChanged                | Integer8             | 1     | 0x656                                                                       |
| uSNCreated                | Integer8             | 1     | 0x656                                                                       |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                           |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                           |

综上所述，computer 类默认的强制继承属性有如下：

- 来自 top 类的 objectClass
- 来自 top 类的 objectCategory
- 来自 top 类的 nTSecurityDescriptor
- 来自 top 类的 instanceType
- 来自 person 类的 cn
- 来自 mailRecipient 类的 cn
- 来自 securityPrincipal 类的 objectSid
- 来自 securityPrincipal 类的 sAMAccountName

这正好与我们通过 Active DirectorySchema 查询的相符合，如图所示。



| 名称                   | 类型 | 系统 | 描述                     | 源类                |
|----------------------|----|----|------------------------|-------------------|
| sAMAccountName       | 强制 | 是  | SAM-Account-Name       | securityPrincipal |
| objectSid            | 强制 | 是  | Object-Sid             | securityPrincipal |
| cn                   | 强制 | 是  | Common-Name            | mailRecipient     |
| cn                   | 强制 | 是  | Common-Name            | person            |
| objectClass          | 强制 | 是  | Object-Class           | top               |
| objectCategory       | 强制 | 是  | Object-Category        | top               |
| nTSecurityDescriptor | 强制 | 是  | NT-Security-Descriptor | top               |
| instanceType         | 强制 | 是  | Instance-Type          | top               |

### 3. Schema Directory Partition 中的类

活动目录中所有的类都在 Schema Directory Partition 内定义了，所有类都是 classSchema 的实例。

如图所示，是 class-Schema 的属性。

| Attribute                 | Syntax               | Count | Value(s)                                                                                                              |
|---------------------------|----------------------|-------|-----------------------------------------------------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | Class-Schema                                                                                                          |
| adminDisplayName          | DirectoryString      | 1     | Class-Schema                                                                                                          |
| cn                        | DirectoryString      | 1     | Class-Schema                                                                                                          |
| defaultHidingValue        | Boolean              | 1     | TRUE                                                                                                                  |
| defaultObjectCategory     | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                              |
| defaultSecurityDescriptor | DirectoryString      | 1     | D:S:                                                                                                                  |
| distinguishedName         | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                              |
| dSCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                                      |
| governsID                 | OID                  | 1     | 1.2.840.113556.1.3.13                                                                                                 |
| instanceType              | Integer              | 1     | 4                                                                                                                     |
| IDAPDisplayName           | DirectoryString      | 1     | classSchema                                                                                                           |
| name                      | DirectoryString      | 1     | Class-Schema                                                                                                          |
| nTSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIID:L:CRPLORC;;;AU)(A;CIID:CCLCSWRPWPLOCRRCWDWO;;;SA)(A;CIID:CCDCLCSWRPWPDTLOCRSDRCWDWO;;;SY)                 |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                              |
| objectClass               | OID                  | 2     | top;classSchema                                                                                                       |
| objectClassCategory       | Enumeration          | 1     | 1                                                                                                                     |
| objectGUID                | OctetString          | 1     | {A5F7BF25-B124-4E86-AC99-8967578D44A0}                                                                                |
| rDNAttID                  | OID                  | 1     | cn                                                                                                                    |
| schemaIDGUID              | OctetString          | 1     | {BF967A83-0DE6-11D0-A285-00AA003049E2}                                                                                |
| showInAdvancedViewO...    | Boolean              | 1     | TRUE                                                                                                                  |
| subClassOf                | OID                  | 1     | top                                                                                                                   |
| systemFlags               | Integer              | 1     | 134217744                                                                                                             |
| systemMayContain          | OID                  | 18    | systemPossSuperiors;systemOnly;systemMustContain;systemMayContain;systemAuxiliaryClass;schemaFlagsEx;rDNAttID;possSup |
| systemMustContain         | OID                  | 6     | subClassOf;schemaIDGUID;objectClassCategory;governsID;defaultObjectCategory;cn                                        |
| systemOnly                | Boolean              | 1     | FALSE                                                                                                                 |
| systemPossSuperiors       | OID                  | 1     | dMD                                                                                                                   |
| uSNCreated                | Integer8             | 1     | 0x5E3                                                                                                                 |
| uSNCreated                | Integer8             | 1     | 0x5E3                                                                                                                 |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:36                                                                                                     |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:36                                                                                                     |

活动目录中的类有三种类型，分别如下：

- 结构类(Structural)：结构类规定了对象实例的基本属性，每个条目属于且仅属于一个结构对象类。前面说过域内每个条目都是类的实例，这个类必须



是结构类，结构类是唯一可以在目录中有实例的类，结构类可以派生于抽象类或其他结构类，可以在结构类定义中包含任意数量的辅助类。

- 抽象类(Abstract)：抽象类是用于导生新结构类的模板，他没有实例，只能充当结构类或者抽象类的父类。一个新的抽象类可以从一个现有的抽象类中派生出来，它只为子类提供属性，将对象属性中公共的部分组织在一起。跟面对对象里面的抽象方法一样，比如说 top 类。
- 辅助类(Auxiliary)：辅助类包含一个属性列表，在结构类或抽象类的定义中添加一个辅助类会将辅助类属性添加到实例中。虽然不能实例化辅助类，但是可以从现有的辅助类或抽象类派生出新的辅助类。例如，Security-Principal 类是一个辅助类，它从名为 Top 的父抽象类中派生。虽然不能实例化辅助类，但可以创建一个结构类 User 的对象，该用户将 Security-Principal 类作为辅助类。因此，该用户对象将拥有辅助类 Security-Principal 的属性，Security-Principal 类的属性可以帮助系统将用户对象识别为安全帐户。辅助类规定了对象实体的扩展属性。虽然每个条目只属于一个结构对象类，但可以同时属于多个辅助对象类。

注：其实还有个 88 类，88 类不属于上述任何一个类别。88 类可以用作抽象类、结构类或辅助类，88 类的类可以继承自所有类。

条目的 objectClassCategory 属性的 count 为 1 说明他是一个结构类，为 2 说明他是一个抽象类，为 3 说明他是一个辅助类。

如图所示，Computer 类的 objectClassCategory 的属性为 1，说明他是一个结构类。而 Computer 类的实例就是域内的机器。

| Attribute                 | Syntax               | Count | Value(s)                                                                                                          |
|---------------------------|----------------------|-------|-------------------------------------------------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | Computer                                                                                                          |
| adminDisplayName          | DirectoryString      | 1     | Computer                                                                                                          |
| auxiliaryClass            | OID                  | 1     | ipHost                                                                                                            |
| cn                        | DirectoryString      | 1     | Computer                                                                                                          |
| defaultHidingValue        | Boolean              | 1     | FALSE                                                                                                             |
| defaultObjectCategory     | DN                   | 1     | CN=Computer,CN=Schema,CN=Configuration,DC=xie,DC=com                                                              |
| defaultSecurityDescriptor | DirectoryString      | 1     | D:({A;;RPWPCRCDDCLCLORCWOWDSDDTSW;;;DA})(A;;RPWPCRCDDCLCLORCWOWDSDDTSW;;;AO)(A;;RPWPCRCDDCLCLORCWOWDSDDTSW;;;SY)  |
| distinguishedName         | DN                   | 1     | CN=Computer,CN=Schema,CN=Configuration,DC=xie,DC=com                                                              |
| dSCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                                  |
| governsID                 | OID                  | 1     | 1.2.840.113556.1.3.30                                                                                             |
| instanceType              | Integer              | 1     | 4                                                                                                                 |
| IDAPDisplayName           | DirectoryString      | 1     | computer                                                                                                          |
| mayContain                | OID                  | 4     | msSFU30Name;nisMapName;msSFU30NisDomain;msSFU30Aliases                                                            |
| name                      | DirectoryString      | 1     | Computer                                                                                                          |
| nTSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIIID;LCRPLORC;;;AU)(A;CIIID;CCLCSWRPWPLOCRRRCWDWO;;;SA)(A;CIIID;CCDCLCSWRPWPOTLOCRSDRCWDWO;;;SY)          |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                          |
| objectClass               | OID                  | 2     | top;classSchema                                                                                                   |
| objectClassCategory       | Enumeration          | 1     | 1                                                                                                                 |
| objectGUID                | OctetString          | 1     | {4751B44D-C567-47C2-98DE-6DD84CC33256}                                                                            |
| rDNAttID                  | OID                  | 1     | cn                                                                                                                |
| schemaIDGUID              | OctetString          | 1     | {BF967A86-0DE6-11D0-A285-00AA003049E2}                                                                            |
| showInAdvancedViewOnly    | Boolean              | 1     | TRUE                                                                                                              |
| subClassOf                | OID                  | 1     | user                                                                                                              |
| systemFlags               | Integer              | 1     | 16                                                                                                                |
| systemMayContain          | OID                  | 51    | msImaging-HashAlgorithm;msImaging-ThumbprintHash;msDS-GenerationId;msTPM-TpmInformationForComputer;msTSSecondaryD |
| systemOnly                | Boolean              | 1     | FALSE                                                                                                             |
| systemPossSuperiors       | OID                  | 3     | container;organizationalUnit;domainDNS                                                                            |
| uSNCreated                | Integer8             | 1     | 0x5E9                                                                                                             |
| uSNCreated                | Integer8             | 1     | 0x5E9                                                                                                             |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:36                                                                                                 |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:36                                                                                                 |

如图所示，Top 类的 objectClassCategory 的属性为 2，说明他是一个抽象类。

| Attribute                 | Syntax               | Count | Value(s)                                                                                                            |
|---------------------------|----------------------|-------|---------------------------------------------------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | Top                                                                                                                 |
| adminDisplayName          | DirectoryString      | 1     | Top                                                                                                                 |
| cn                        | DirectoryString      | 1     | Top                                                                                                                 |
| defaultHidingValue        | Boolean              | 1     | TRUE                                                                                                                |
| defaultObjectCategory     | DN                   | 1     | CN=Top,CN=Schema,CN=Configuration,DC=xie,DC=com                                                                     |
| defaultSecurityDescriptor | DirectoryString      | 1     | D:({A;;RPWPCRCDDCLCLORCWOWDSDDTSW;;;DA})(A;;RPWPCRCDDCLCLORCWOWDSDDTSW;;;SY)(A;;RPLCLORC;;;AU)                      |
| distinguishedName         | DN                   | 1     | CN=Top,CN=Schema,CN=Configuration,DC=xie,DC=com                                                                     |
| dSCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                                    |
| governsID                 | OID                  | 1     | 2.5.6.0                                                                                                             |
| instanceType              | Integer              | 1     | 4                                                                                                                   |
| IDAPDisplayName           | DirectoryString      | 1     | top                                                                                                                 |
| mayContain                | OID                  | 4     | msSFU30PosixMemberOf;msDFSR-ComputerReferenceBL;msDFSR-MemberReferenceBL;msDS-ObjectReferenceBL                     |
| name                      | DirectoryString      | 1     | Top                                                                                                                 |
| nTSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIIID;LCRPLORC;;;AU)(A;CIIID;CCLCSWRPWPLOCRRRCWDWO;;;SA)(A;CIIID;CCDCLCSWRPWPOTLOCRSDRCWDWO;;;SY)            |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                            |
| objectClass               | OID                  | 2     | top;classSchema                                                                                                     |
| objectClassCategory       | Enumeration          | 1     | 2                                                                                                                   |
| objectGUID                | OctetString          | 1     | {4C745911-D477-4C51-A9AC-9EA28AFD5765}                                                                              |
| rDNAttID                  | OID                  | 1     | cn                                                                                                                  |
| schemaIDGUID              | OctetString          | 1     | {BF967A87-0DE6-11D0-A285-00AA003049E2}                                                                              |
| showInAdvancedViewOnly    | Boolean              | 1     | TRUE                                                                                                                |
| subClassOf                | OID                  | 1     | top                                                                                                                 |
| systemFlags               | Integer              | 1     | 16                                                                                                                  |
| systemMayContain          | OID                  | 114   | msDS-RepValueMetaDataType;msDS-parentDistname;msds-memberTransitive;msds-memberOfTransitive;msDS-TDOEGressBL;msDS-T |
| systemMustContain         | OID                  | 4     | objectClass;objectCategory;nTSecurityDescriptor;instanceType                                                        |
| systemOnly                | Boolean              | 1     | TRUE                                                                                                                |
| systemPossSuperiors       | OID                  | 1     | lostAndFound                                                                                                        |
| uSNCreated                | Integer8             | 1     | 0x751                                                                                                               |
| uSNCreated                | Integer8             | 1     | 0x751                                                                                                               |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                                   |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                                   |

如图所示，Security-Principal 类的 objectClassCategory 的属性为 3，说明他是一个辅助类。

| Attribute              | Syntax               | Count | Value(s)                                                                                                                    |
|------------------------|----------------------|-------|-----------------------------------------------------------------------------------------------------------------------------|
| adminDescription       | DirectoryString      | 1     | Security-Principal                                                                                                          |
| adminDisplayName       | DirectoryString      | 1     | Security-Principal                                                                                                          |
| cn                     | DirectoryString      | 1     | Security-Principal                                                                                                          |
| defaultHidingValue     | Boolean              | 1     | TRUE                                                                                                                        |
| defaultObjectCategory  | DN                   | 1     | CN=Security-Principal,CN=Schema,CN=Configuration,DC=xie,DC=com                                                              |
| distinguishedName      | DN                   | 1     | CN=Security-Principal,CN=Schema,CN=Configuration,DC=xie,DC=com                                                              |
| dSCorePropagationData  | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                                            |
| governorID             | OID                  | 1     | 1.2.840.113556.1.5.6                                                                                                        |
| instanceType           | Integer              | 1     | 4                                                                                                                           |
| IDAPDisplayName        | DirectoryString      | 1     | securityPrincipal                                                                                                           |
| name                   | DirectoryString      | 1     | Security-Principal                                                                                                          |
| nTSecurityDescriptor   | NTSecurityDescriptor | 1     | D:AI(A;CIID;LCRPLORC;;;AU)(A;CIID;CCLCSWRPWLOCRCWDWO;;;SA)(A;CIID;CCDCLCSWRPWPDTLORSDRCWDWO;;;SY)                           |
| objectCategory         | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                                    |
| objectClass            | OID                  | 2     | top:classSchema                                                                                                             |
| objectClassCategory    | Enumeration          | 1     | 3                                                                                                                           |
| objectGUID             | OctetString          | 1     | {E6D751FE-E00C-40DA-9098-AC06C7EFE2F7}                                                                                      |
| rDNAttID               | OID                  | 1     | cn                                                                                                                          |
| schemaIDGUID           | OctetString          | 1     | {BF967AB0-0DE6-11D0-A285-00AA003049E2}                                                                                      |
| showInAdvancedViewO... | Boolean              | 1     | TRUE                                                                                                                        |
| subClassOf             | OID                  | 1     | top                                                                                                                         |
| systemFlags            | Integer              | 1     | 16                                                                                                                          |
| systemMayContain       | OID                  | 12    | supplementalCredentials;sidHistory;securityIdentifier;sAMAccountType;rid;tokenGroupsNoGCAcceptable;tokenGroupsGlobalAndL... |
| systemMustContain      | OID                  | 2     | sAMAccountName;objectSid                                                                                                    |
| systemOnly             | Boolean              | 1     | FALSE                                                                                                                       |
| uSNCreated             | Integer8             | 1     | 0x732                                                                                                                       |
| uSNChanged             | Integer8             | 1     | 0x732                                                                                                                       |
| whenChanged            | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                                           |
| whenCreated            | GeneralizedTime      | 1     | 2013/5/22 0:44:37                                                                                                           |

#### 4. Schema Directory Partition 中的属性

Schema Directory Partition 中除了定义了 Active Directory 中使用的类之外，还定义了 Active Directory 中使用的属性。每个属性在 Schema Directory Partition 内也是一个条目，是类 Attribute-Schema 的实例。

如图所示，查看 Attribute-Schema 的属性。

| Attribute                 | Syntax               | Count | Value(s)                                                                                                       |
|---------------------------|----------------------|-------|----------------------------------------------------------------------------------------------------------------|
| adminDescription          | DirectoryString      | 1     | Attribute-Schema                                                                                               |
| adminDisplayName          | DirectoryString      | 1     | Attribute-Schema                                                                                               |
| cn                        | DirectoryString      | 1     | Attribute-Schema                                                                                               |
| defaultHidingValue        | Boolean              | 1     | TRUE                                                                                                           |
| defaultObjectCategory     | DN                   | 1     | CN=Attribute-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                   |
| defaultSecurityDescriptor | DirectoryString      | 1     | D:S:                                                                                                           |
| distinguishedName         | DN                   | 1     | CN=Attribute-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                   |
| dSCorePropagationData     | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                               |
| governsID                 | OID                  | 1     | 1.2.840.113556.1.3.14                                                                                          |
| instanceType              | Integer              | 1     | 4                                                                                                              |
| DAPDisplayName            | DirectoryString      | 1     | attributeSchema                                                                                                |
| name                      | DirectoryString      | 1     | Attribute-Schema                                                                                               |
| nTSecurityDescriptor      | NTSecurityDescriptor | 1     | D:AI(A;CIID;LCRPLORC;;;AU)(A;CIID;CCLCSWRPWLOCRCWDWO;;;SA)(A;CIID;CCDCLCSWRPWPDLOCRCWDWO;;;)                   |
| objectCategory            | DN                   | 1     | CN=Class-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                                       |
| objectClass               | OID                  | 2     | top;classSchema                                                                                                |
| objectClassCategory       | Enumeration          | 1     | 1                                                                                                              |
| objectGUID                | OctetString          | 1     | {C1C44649-E964-4238-BFDD-512420CE1264}                                                                         |
| rDNAttID                  | OID                  | 1     | cn                                                                                                             |
| schemaIDGUID              | OctetString          | 1     | {BF967A80-0DE6-11D0-A285-00AA003049E2}                                                                         |
| showInAdvancedViewO...    | Boolean              | 1     | TRUE                                                                                                           |
| subClassOf                | OID                  | 1     | top                                                                                                            |
| systemFlags               | Integer              | 1     | 134217744                                                                                                      |
| systemMayContain          | OID                  | 16    | systemOnly;searchFlags;schemaFlagsEx;rangeUpper;rangeLower;oMObjectClass;msDs-Schema-Extensions;msDS-IntId;mAP |
| systemMustContain         | OID                  | 7     | schemaIDGUID;oMSyntax;DAPDisplayName;isSingleValued;cn;attributeSyntax;attributeID                             |
| systemOnly                | Boolean              | 1     | FALSE                                                                                                          |
| systemPossSuperiors       | OID                  | 1     | dMD                                                                                                            |
| uSNChanged                | Integer8             | 1     | 0x5DA                                                                                                          |
| uSNCreated                | Integer8             | 1     | 0x5DA                                                                                                          |
| whenChanged               | GeneralizedTime      | 1     | 2013/5/22 0:44:36                                                                                              |
| whenCreated               | GeneralizedTime      | 1     | 2013/5/22 0:44:36                                                                                              |

在域内的所有属性必须在这里定义，而这里的条目，最主要的是限定了属性的语法定义，其实就是数据类型。属性的值的数据类型由属性条目的 attributeSyntax 属性和 oMSyntax 属性决定。这听起来有点绕口，有点像先有鸡还是先有蛋的感觉，我们往下看。

如图所示，以属性 Object-Sid (CN=Object-Sid,CN=Schema,CN=Configuration,DC=xie,DC=com)为例。

Path: CN=Object-Sid,CN=Schema,CN=Configuration,DC=xie,DC=com,10.211.55.4 [DC.xie.com]

| Attribute                  | Syntax               | Count | Value(s)                                                                                    |
|----------------------------|----------------------|-------|---------------------------------------------------------------------------------------------|
| adminDescription           | DirectoryString      | 1     | Object-Sid                                                                                  |
| adminDisplayName           | DirectoryString      | 1     | Object-Sid                                                                                  |
| attributeID                | OID                  | 1     | 1.2.840.113556.1.4.146                                                                      |
| attributeSecurityGUID      | OctetString          | 1     | {598A2F42-79A2-11D0-9020-00C0#C2D3CF}                                                       |
| attributeSyntax            | OID                  | 1     | 2.5.5.17                                                                                    |
| cn                         | DirectoryString      | 1     | Object-Sid                                                                                  |
| distinguishedName          | DN                   | 1     | CN=Object-Sid,CN=Schema,CN=Configuration,DC=xie,DC=com                                      |
| dSCorePropagationData      | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                            |
| instanceType               | Integer              | 1     | 4                                                                                           |
| isMemberOfPartialAttrib... | Boolean              | 1     | TRUE                                                                                        |
| isSingleValued             | Boolean              | 1     | TRUE                                                                                        |
| LDAPDisplayName            | DirectoryString      | 1     | Object-Sid                                                                                  |
| APIID                      | Integer              | 1     | 32807                                                                                       |
| name                       | DirectoryString      | 1     | Object-Sid                                                                                  |
| NTSecurityDescriptor       | NTSecurityDescriptor | 1     | D:AI(A;CIID)LCRPLORC:::AUJ(A;CIID)CCLCSWRPWLOCRCWDWO:::SA(A;CIID)CCDCLCSWRPWDTLOCRSORDCWDWO |
| objectCategory             | DN                   | 1     | CN=Attribute-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                |
| objectClass                | OID                  | 2     | top,attributeSchema                                                                         |
| objectGUID                 | OctetString          | 1     | {37625A89-576C-49D0-AFCA-6353A1CE19F8}                                                      |
| oMSyntax                   | Integer              | 1     | 4                                                                                           |
| rangeLower                 | Integer              | 1     | 0                                                                                           |
| rangeUpper                 | Integer              | 1     | 28                                                                                          |
| schemaFlagsEx              | Integer              | 1     | 1                                                                                           |
| schemaIDGUID               | OctetString          | 1     | {BF9679E8-0DE6-11D0-A285-00AA003049E2}                                                      |
| searchFlags                | Enumeration          | 1     | 9                                                                                           |
| showInAdvancedViewO...     | Boolean              | 1     | TRUE                                                                                        |
| systemFlags                | Integer              | 1     | 18                                                                                          |
| systemOnly                 | Boolean              | 1     | TRUE                                                                                        |
| uSNChanged                 | Integer8             | 1     | 0x373                                                                                       |
| uSNCreated                 | Integer8             | 1     | 0x373                                                                                       |
| whenChanged                | GeneralizedTime      | 1     | 2013/5/22 0:44:35                                                                           |
| whenCreated                | GeneralizedTime      | 1     | 2013/5/22 0:44:35                                                                           |

他的 attributeSyntax 属性是 2.5.5.17，如图所示。

### Attribute Properties

Attribute: attributeSyntax

Object: CN=Object-Sid,CN=Schema,CN=Configuration,DC=xie,DC=com

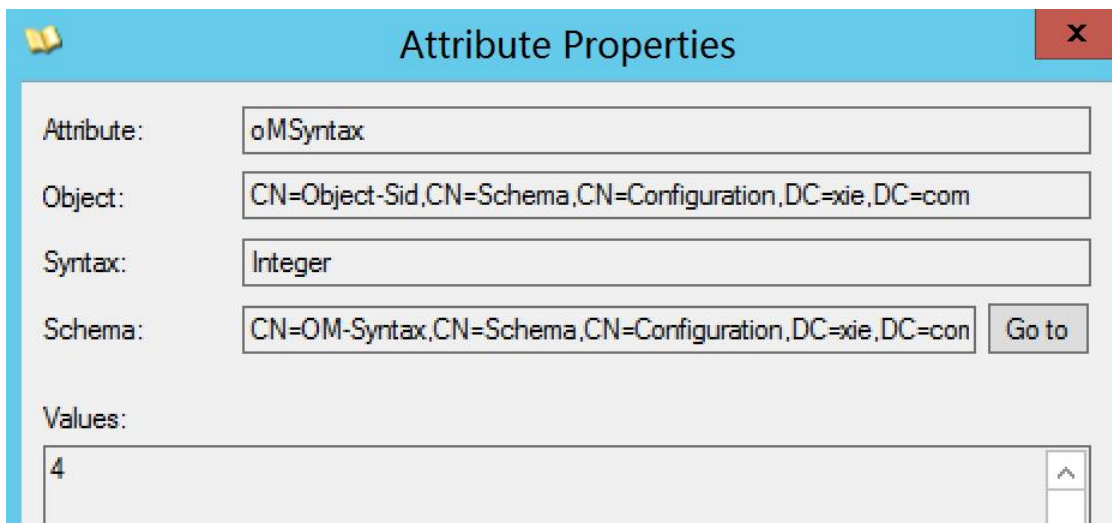
Syntax: OID

Schema: CN=Attribute-Syntax,CN=Schema,CN=Configuration,DC=xie,DC Go to

Values: 2.5.5.17

oMSyntax 属性是 4，如图所示。





如图所示，通过查表可以得知 Object-Sid 属性的值是一个安全标识符(SID)，因此 Object-Sid 属性的值的数据类型为 SID。

|             |          |   |          |                           |
|-------------|----------|---|----------|---------------------------|
| String(Sid) | 2.5.5.17 | 4 | \x550511 | Security identifier (SID) |
|-------------|----------|---|----------|---------------------------|

如图所示，查看 hack 对象的 objectsid 属性的值的数据类型，可以看到确实是 sid。

| Attribute             | Syntax               | Count | Value(s)                                                                                                  |
|-----------------------|----------------------|-------|-----------------------------------------------------------------------------------------------------------|
| accountExpires        | Integer8             | 1     | 0x0                                                                                                       |
| badPasswordTime       | Integer8             | 1     | 0x0                                                                                                       |
| badPwdCount           | Integer              | 1     | 0                                                                                                         |
| cn                    | DirectoryString      | 1     | hack                                                                                                      |
| codePage              | Integer              | 1     | 0                                                                                                         |
| countryCode           | Integer              | 1     | 0                                                                                                         |
| distinguishedName     | DN                   | 1     | CN=hack,CN=Users,DC=xie,DC=com                                                                            |
| dsCorePropagationData | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                          |
| instanceType          | Integer              | 1     | 4                                                                                                         |
| lastLogoff            | Integer8             | 1     | 0x0                                                                                                       |
| lastLogon             | Integer8             | 1     | 2021/10/26 22:12:25                                                                                       |
| lastLogonTimestamp    | Integer8             | 1     | 2021/11/6 9:56:10                                                                                         |
| logonCount            | Integer              | 1     | 11                                                                                                        |
| name                  | DirectoryString      | 1     | hack                                                                                                      |
| ntSecurityDescriptor  | NTSecurityDescriptor | 1     | D:AI(OA;;RP;4c164200-20c0-11d0-a768-00aa006e0529;RS)(OA;;RP;5f202010-79a5-11d0-9020-00c04fc2d4cf;RS)(OA;; |
| objectCategory        | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                                                        |
| objectClass           | OID                  | 4     | top;person;organizationalPerson;user                                                                      |
| objectGUID            | OctetString          | 1     | {908A4F43-E2C4-4994-8728-B1C42507D357}                                                                    |
| objectSid             | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-1105                                                            |
| primaryGroupID        | Integer              | 1     | 513                                                                                                       |
| pwdLastSet            | Integer8             | 1     | 2021/10/16 11:37:28                                                                                       |
| sAMAccountName        | DirectoryString      | 1     | hack                                                                                                      |
| sAMAccountType        | Integer              | 1     | 805306368                                                                                                 |
| userAccountControl    | Integer              | 1     | 512                                                                                                       |
| uSNChanged            | Integer8             | 1     | 0x5065                                                                                                    |
| uSNCreated            | Integer8             | 1     | 0x31E5                                                                                                    |
| whenChanged           | GeneralizedTime      | 1     | 2021/11/6 9:56:10                                                                                         |
| whenCreated           | GeneralizedTime      | 1     | 2021/10/16 11:37:28                                                                                       |

而这里的 Object-Sid 属性条目的 attributeSyntax 属性和 oMSyntax 属性又可以在 Schema Directory Partition 中找到定义。而 attributeSyntax 属性条目和 oMSyntax 属性条目自身又是通过他们自身的 attributeSyntax 属性和 oMSyntax 属性来定义数据类型的。

如图所示，查看条目 attribute-Syntax 的 attributeSyntax 属性和 oMSyntax 属性。

|                                                                                             |                      |       |                                                                                                       |
|---------------------------------------------------------------------------------------------|----------------------|-------|-------------------------------------------------------------------------------------------------------|
| Path: CN=Attribute-Syntax,CN=Schema,CN=Configuration,DC=xie,DC=com,10.211.55.4 [DC.xie.com] |                      |       |                                                                                                       |
| Attribute                                                                                   | Syntax               | Count | Value(s)                                                                                              |
| adminDescription                                                                            | DirectoryString      | 1     | Attribute-Syntax                                                                                      |
| adminDisplayName                                                                            | DirectoryString      | 1     | Attribute-Syntax                                                                                      |
| attributeID                                                                                 | OID                  | 1     | 1.2.840.113556.1.2.32                                                                                 |
| attributeSyntax                                                                             | OID                  | 1     | 2.5.5.2                                                                                               |
| cn                                                                                          | DirectoryString      | 1     | Attribute-Syntax                                                                                      |
| distinguishedName                                                                           | DN                   | 1     | CN=Attribute-Syntax,CN=Schema,CN=Configuration,DC=xie,DC=com                                          |
| dsCorePropagationData                                                                       | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                      |
| instanceType                                                                                | Integer              | 1     | 4                                                                                                     |
| isSingleValued                                                                              | Boolean              | 1     | TRUE                                                                                                  |
| LDAPDisplayName                                                                             | DirectoryString      | 1     | Attribute-Syntax                                                                                      |
| name                                                                                        | DirectoryString      | 1     | Attribute-Syntax                                                                                      |
| ntSecurityDescriptor                                                                        | NTSecurityDescriptor | 1     | D:AI(A;CIID;L;CRP;ORC;);AUJ(A;CIID;CCLCS;WRPWPLOCRRC;WDWO;);SA(A;CIID;CCDCLCS;WRPWPDTLOCRDR;C;WDWO;); |
| objectCategory                                                                              | DN                   | 1     | CN=Attribute-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                          |
| objectClass                                                                                 | OID                  | 2     | top;attributeSchema                                                                                   |
| objectGUID                                                                                  | OctetString          | 1     | {D0E1588E-8DE4-4A2E-A2C7-4C6D5E3BA4F6}                                                                |
| oMSyntax                                                                                    | Integer              | 1     | 6                                                                                                     |
| schemaFlagsEx                                                                               | Integer              | 1     | 1                                                                                                     |
| schemaIDGUID                                                                                | OctetString          | 1     | {BF967925-0DE6-11D0-A285-00AA003049E2}                                                                |
| searchFlags                                                                                 | Enumeration          | 1     | 8                                                                                                     |
| showInAdvancedViewO...                                                                      | Boolean              | 1     | TRUE                                                                                                  |
| systemFlags                                                                                 | Integer              | 1     | 16                                                                                                    |
| systemOnly                                                                                  | Boolean              | 1     | TRUE                                                                                                  |
| uSNChanged                                                                                  | Integer8             | 1     | 0x52                                                                                                  |
| uSNCreated                                                                                  | Integer8             | 1     | 0x52                                                                                                  |
| whenChanged                                                                                 | GeneralizedTime      | 1     | 2013/5/22 0:44:33                                                                                     |
| whenCreated                                                                                 | GeneralizedTime      | 1     | 2013/5/22 0:44:33                                                                                     |

如图所示，查看条目 oM-Syntax 的 attributeSyntax 属性和 oMSyntax 属性。

Path: CN=OM-Syntax,CN=Schema,CN=Configuration,DC=xie,DC=com,10.211.55.4 [DC.xie.com]

| Attribute              | Syntax               | Count | Value(s)                                                                                    |
|------------------------|----------------------|-------|---------------------------------------------------------------------------------------------|
| adminDescription       | DirectoryString      | 1     | OM-Syntax                                                                                   |
| adminDisplayName       | DirectoryString      | 1     | OM-Syntax                                                                                   |
| attributeSyntax        | OID                  | 1     | 2.5.5.9                                                                                     |
| cn                     | DirectoryString      | 1     | OM-Syntax                                                                                   |
| distinguishedName      | DN                   | 1     | CN=OM-Syntax,CN=Schema,CN=Configuration,DC=xie,DC=com                                       |
| dsCorePropagationData  | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                            |
| instanceType           | Integer              | 1     | 4                                                                                           |
| isSingleValued         | Boolean              | 1     | TRUE                                                                                        |
| IDAPDisplayName        | DirectoryString      | 1     | oMSyntax                                                                                    |
| mAPIID                 | Integer              | 1     | 33022                                                                                       |
| name                   | DirectoryString      | 1     | OM-Syntax                                                                                   |
| ntSecurityDescriptor   | NTSecurityDescriptor | 1     | D:AI(A;CIID;LCRPLORC;;AU)(A;CIID;CCLCSWRPWLOCRRCWDWO;;SA)(A;CIID;CCDCLCSWRPWLOCRRCWDWO;;SA) |
| objectCategory         | DN                   | 1     | CN=Attribute-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                |
| objectClass            | OID                  | 2     | top,attributeSchema                                                                         |
| objectGUID             | OctetString          | 1     | {A8C75A1E-5206-44A3-BE75-B13CD742C560}                                                      |
| oMSyntax               | Integer              | 1     | 2                                                                                           |
| schemaFlagsEx          | Integer              | 1     | 1                                                                                           |
| schemaIDGUID           | OctetString          | 1     | {BF9679ED-00E6-11D0-A285-00AA003049E2}                                                      |
| searchFlags            | Enumeration          | 1     | 8                                                                                           |
| showInAdvancedViewOnly | Boolean              | 1     | TRUE                                                                                        |
| systemFlags            | Integer              | 1     | 16                                                                                          |
| systemOnly             | Boolean              | 1     | TRUE                                                                                        |
| usnChanged             | Integer8             | 1     | 0x377                                                                                       |
| usnCreated             | Integer8             | 1     | 0x377                                                                                       |
| whenChanged            | GeneralizedTime      | 1     | 2013/5/22 0:44:35                                                                           |
| whenCreated            | GeneralizedTime      | 1     | 2013/5/22 0:44:35                                                                           |

## Application Directory Partition

从 Windows Server 2003 开始，微软引入了应用程序目录分区(Application Directory Partition)，其允许用户自定义分区来扩展目录分区。它提供了控制复制范围的能力，并允许以更适合动态数据的方式放置副本。应用程序目录分区会被复制到林中特定的域控中，而不是所有的域控中。管理员可以创建应用程序目录分区，以将数据存储在他们选择的特定域控制器上。

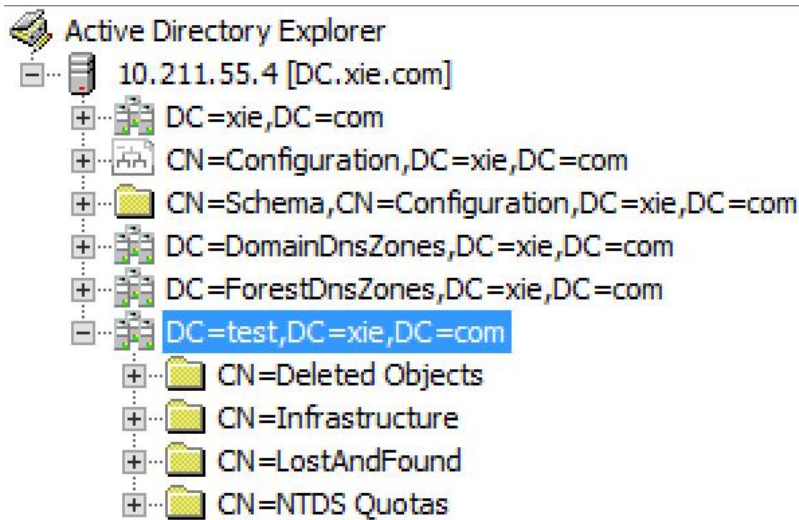
Application Directory Partitions 主要有以下特点：

- 如果用户想要定义一个分区，可以通过 Application Directory Partitions。虽然微软也预置了两个 Application Directory Partitions，但是 Application Directory Partitions 的设计更多是为了让用户可以自定义自己的数据。设计 Application Partitions 最大的用途就是，让用户自己来定义分区。
- Application Directory Partitions 可以存储动态对象。动态对象是具有生存时间(TTL)值的对象，该值确定它们在被 Active Directory 自动删除之前将存在多长时间。也就是说 Application Directory Partitions 可以给数据设置个 TTL，时间一到，Active Directory 就删除该数据。

如图所示，通过 ntdsutil 创建应用程序目录分区 DC=test,DC=xie,DC=com。

```
C:\Users\Administrator>ntdsutil
ntdsutil: partition management
partition management: create nc "dc=test,dc=xie,dc=com" dc.xie.com
adding object dc=test,dc=xie,dc=com
partition management:
```

如图所示，使用 ADEplorer 连接活动目录查看应用程序目录分区，可以看到应用程序目录分区 DC=test,DC=xie,DC=com。



## 条目属性分析

我们下面具体分析一下条目的一些比较重要的属性。首先看看条目属性的四列分别代表着什么？

- Attribute 代表条目的属性。
- Syntna 代表条目属性的值的数据类型。
- Count 代表条目属性的值的个数。







| Attribute              | Syntax               | Count | Value(s)                                                                         |
|------------------------|----------------------|-------|----------------------------------------------------------------------------------|
| accountExpires         | Integer8             | 1     | 0x0                                                                              |
| adminCount             | Integer              | 1     | 1                                                                                |
| badPasswordTime        | Integer8             | 1     | 2021/10/16 11:29:34                                                              |
| badPwdCount            | Integer              | 1     | 0                                                                                |
| cn                     | DirectoryString      | 1     | Administrator                                                                    |
| codePage               | Integer              | 1     | 0                                                                                |
| countryCode            | Integer              | 1     | 0                                                                                |
| description            | DirectoryString      | 1     | 管理计算机(域)的内置帐户                                                                    |
| distinguishedName      | DN                   | 1     | CN=Administrator,CN=Users,DC=xie,DC=com                                          |
| dSCorePropagationData  | GeneralizedTime      | 4     | 2021/10/16 11:42:05;2021/10/16 11:42:05;2021/10/16 11:26:55;1601/1/2 1:12:16     |
| instanceType           | Integer              | 1     | 4                                                                                |
| isCriticalSystemObject | Boolean              | 1     | TRUE                                                                             |
| lastLogoff             | Integer8             | 1     | 0x0                                                                              |
| lastLogon              | Integer8             | 1     | 2021/10/26 22:12:01                                                              |
| lastLogonTimestamp     | Integer8             | 1     | 2021/10/16 11:29:51                                                              |
| logonCount             | Integer              | 1     | 20                                                                               |
| logonHours             | OctetString          | 1     | 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255                  |
| memberOf               | DN                   | 5     | CN=Group Policy Creator Owners,CN=Users,DC=xie,DC=com;CN=Domain Admins,CN=Users  |
| name                   | DirectoryString      | 1     | Administrator                                                                    |
| nTSecurityDescriptor   | NTSecurityDescriptor | 1     | D:PAI(OA;;RP;4c164200-20c0-11d0-a768-00aa006e0529;4828cc14-1437-45bc-9b07-ad6f01 |
| objectCategory         | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                               |
| objectClass            | OID                  | 4     | top;person;organizationalPerson;user                                             |
| objectGUID             | OctetString          | 1     | {7B73B1AA-7830-48F7-9106-0BE3FA003BCE}                                           |
| objectSid              | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-500                                    |
| primaryGroupID         | Integer              | 1     | 513                                                                              |
| pwdLastSet             | Integer8             | 1     | 2021/10/16 11:29:51                                                              |
| sAMAccountName         | DirectoryString      | 1     | Administrator                                                                    |
| sAMAccountType         | Integer              | 1     | 805306368                                                                        |
| userAccountControl     | Integer              | 1     | 512                                                                              |
| uSNChanged             | Integer8             | 1     | 0x31F3                                                                           |
| uSNCreated             | Integer8             | 1     | 0x2004                                                                           |
| whenChanged            | GeneralizedTime      | 1     | 2021/10/16 11:42:05                                                              |
| whenCreated            | GeneralizedTime      | 1     | 2021/10/16 11:24:44                                                              |

### 3. badPwdCount

badPwdCount 属性的话很好理解，就是该用户输错密码的次数。其实可以根据这一属性，来查看哪些用户可能被爆破。

如图所示，administrator 的 badPwdCount 为 0，说明没输错过密码。



| Attribute              | Syntax                 | Count    | Value(s)                                                                        |
|------------------------|------------------------|----------|---------------------------------------------------------------------------------|
| accountExpires         | Integer8               | 1        | 0x0                                                                             |
| adminCount             | Integer                | 1        | 1                                                                               |
| badPasswordTime        | Integer8               | 1        | 2021/10/16 11:29:34                                                             |
| badPwdCount            | Integer                | 1        | 0                                                                               |
| <b>cn</b>              | <b>DirectoryString</b> | <b>1</b> | <b>Administrator</b>                                                            |
| codePage               | Integer                | 1        | 0                                                                               |
| countryCode            | Integer                | 1        | 0                                                                               |
| description            | DirectoryString        | 1        | 管理计算机(域)的内置帐户                                                                   |
| distinguishedName      | DN                     | 1        | CN=Administrator,CN=Users,DC=xie,DC=com                                         |
| dsCorePropagationData  | GeneralizedTime        | 4        | 2021/10/16 11:42:05;2021/10/16 11:42:05;2021/10/16 11:42:05;2021/10/16 11:42:05 |
| instanceType           | Integer                | 1        | 4                                                                               |
| isCriticalSystemObject | Boolean                | 1        | TRUE                                                                            |
| lastLogoff             | Integer8               | 1        | 0x0                                                                             |
| lastLogon              | Integer8               | 1        | 2021/10/26 22:12:01                                                             |
| lastLogonTimestamp     | Integer8               | 1        | 2021/10/16 11:29:51                                                             |
| logonCount             | Integer                | 1        | 20                                                                              |
| logonHours             | OctetString            | 1        | 255 255 255 255 255 255 255 255 255 255 255 255                                 |
| memberOf               | DN                     | 5        | CN=Group Policy Creator Owners,CN=Users,DC=xie,DC=com                           |
| name                   | DirectoryString        | 1        | Administrator                                                                   |
| ntSecurityDescriptor   | NTSecurityDescriptor   | 1        | D:PAI(OA;;RP;4c164200-20c0-11d0-a768-00aa006e05b9)                              |
| objectCategory         | DN                     | 1        | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                              |
| objectClass            | OID                    | 4        | top;person;organizationalPerson;user                                            |
| objectGUID             | OctetString            | 1        | {7B73B1AA-7830-48F7-9106-0BE3FA003BCE}                                          |
| objectSid              | Sid                    | 1        | S-1-5-21-1313979556-3624129433-4055459191-500                                   |
| primaryGroupID         | Integer                | 1        | 513                                                                             |
| pwdLastSet             | Integer8               | 1        | 2021/10/16 11:29:51                                                             |
| sAMAccountName         | DirectoryString        | 1        | Administrator                                                                   |
| sAMAccountType         | Integer                | 1        | 805306368                                                                       |
| userAccountControl     | Integer                | 1        | 512                                                                             |
| uSNChanged             | Integer8               | 1        | 0x31F3                                                                          |
| uSNCreated             | Integer8               | 1        | 0x2004                                                                          |
| whenChanged            | GeneralizedTime        | 1        | 2021/10/16 11:42:05                                                             |
| whenCreated            | GeneralizedTime        | 1        | 2021/10/16 11:24:44                                                             |

## 5. description

description 属性的话很好理解，就是对该条目的一个描述。

如图所示，域管理员 administrator 的描述为“管理计算机(域)的内置帐户”。





| Attribute              | Syntax               | Count | Value(s)                                                                            |
|------------------------|----------------------|-------|-------------------------------------------------------------------------------------|
| accountExpires         | Integer8             | 1     | 0x0                                                                                 |
| adminCount             | Integer              | 1     | 1                                                                                   |
| badPasswordTime        | Integer8             | 1     | 2021/10/16 11:29:34                                                                 |
| badPwdCount            | Integer              | 1     | 0                                                                                   |
| cn                     | DirectoryString      | 1     | Administrator                                                                       |
| codePage               | Integer              | 1     | 0                                                                                   |
| countryCode            | Integer              | 1     | 0                                                                                   |
| description            | DirectoryString      | 1     | 管理计算机(域)的内网帐户                                                                       |
| distinguishedName      | DN                   | 1     | CN=Administrator,CN=Users,DC=xie,DC=com                                             |
| dSCorePropagationData  | GeneralizedTime      | 4     | 2021/10/16 11:42:05;2021/10/16 11:42:05;2021/10/16 11:26:55;1601/1/2 1:12:16        |
| instanceType           | Integer              | 1     | 4                                                                                   |
| isCriticalSystemObject | Boolean              | 1     | TRUE                                                                                |
| lastLogoff             | Integer8             | 1     | 0x0                                                                                 |
| lastLogon              | Integer8             | 1     | 2021/10/26 22:12:01                                                                 |
| lastLogonTimestamp     | Integer8             | 1     | 2021/10/16 11:29:51                                                                 |
| logonCount             | Integer              | 1     | 20                                                                                  |
| logonHours             | OctetString          | 1     | 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 |
| memberOf               | DN                   | 5     | CN=Group Policy Creator Owners,CN=Users,DC=xie,DC=com;CN=Domain Admins,CN=U         |
| name                   | DirectoryString      | 1     | Administrator                                                                       |
| nTSecurityDescriptor   | NTSecurityDescriptor | 1     | D:PAI(OA;RP;4c164200-20c0-11d0-a768-00aa006e0529;4828cc14-1437-45bc-9b07-ad6        |
| objectCategory         | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                                  |
| objectClass            | OID                  | 4     | top;person;organizationalPerson;user                                                |
| objectGUID             | OctetString          | 1     | {7B73B1AA-7830-48F7-9106-0BE3FA003BCE}                                              |
| objectSid              | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-500                                       |
| primaryGroupID         | Integer              | 1     | 513                                                                                 |
| pwdLastSet             | Integer8             | 1     | 2021/10/16 11:29:51                                                                 |
| sAMAccountName         | DirectoryString      | 1     | Administrator                                                                       |
| sAMAccountType         | Integer              | 1     | 805306368                                                                           |
| userAccountControl     | Integer              | 1     | 512                                                                                 |
| uSNChanged             | Integer8             | 1     | 0x31F3                                                                              |
| uSNCreated             | Integer8             | 1     | 0x2004                                                                              |
| whenChanged            | GeneralizedTime      | 1     | 2021/10/16 11:42:05                                                                 |
| whenCreated            | GeneralizedTime      | 1     | 2021/10/16 11:24:44                                                                 |

## 7. lastLogon

lastLogon 属性代表该用户最后的登录时间。

如图所示，可以看到 administrator 用户最后一次登录时间为 2021/10/26 22:12:01。







## 11. name

name 属性就是条目的名字。如图所示，administrator 用户的 name 属性的值就为 administrator。

| Attribute              | Syntax               | Count | Value(s)                                                                                      |
|------------------------|----------------------|-------|-----------------------------------------------------------------------------------------------|
| accountExpires         | Integer8             | 1     | 0x0                                                                                           |
| adminCount             | Integer              | 1     | 1                                                                                             |
| badPasswordTime        | Integer8             | 1     | 2021/10/16 11:29:34                                                                           |
| badPwdCount            | Integer              | 1     | 0                                                                                             |
| cn                     | DirectoryString      | 1     | Administrator                                                                                 |
| codePage               | Integer              | 1     | 0                                                                                             |
| countryCode            | Integer              | 1     | 0                                                                                             |
| description            | DirectoryString      | 1     | 管理计算机(域)的内置帐户                                                                                 |
| distinguishedName      | DN                   | 1     | CN=Administrator,CN=Users,DC=xie,DC=com                                                       |
| dSCorePropagationData  | GeneralizedTime      | 4     | 2021/10/16 11:42:05;2021/10/16 11:42:05;2021/10/16 11:26:55;1601/1/2 1:12:16                  |
| instanceType           | Integer              | 1     | 4                                                                                             |
| isCriticalSystemObject | Boolean              | 1     | TRUE                                                                                          |
| lastLogoff             | Integer8             | 1     | 0x0                                                                                           |
| lastLogon              | Integer8             | 1     | 2021/10/26 22:12:01                                                                           |
| lastLogonTimestamp     | Integer8             | 1     | 2021/10/16 11:29:51                                                                           |
| logonCount             | Integer              | 1     | 20                                                                                            |
| logonHours             | OctetString          | 1     | 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255               |
| memberOf               | DN                   | 5     | CN=Group Policy Creator Owners,CN=Users,DC=xie,DC=com;CN=Domain Admins,CN=Users,DC=xie,DC=com |
| name                   | DirectoryString      | 1     | Administrator                                                                                 |
| nTSecurityDescriptor   | NTSecurityDescriptor | 1     | D:PAI(OA;;RP;4c164200-20c0-11d0-a768-00aa006e0529;4828cc14-1437-45bc-9b07-ad6f015e            |
| objectCategory         | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                                            |
| objectClass            | OID                  | 4     | top;person;organizationalPerson;user                                                          |
| objectGUID             | OctetString          | 1     | {7B73B1AA-7830-48F7-9106-0BE3FA003BCE}                                                        |
| objectSid              | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-500                                                 |
| primaryGroupID         | Integer              | 1     | 513                                                                                           |
| pwdLastSet             | Integer8             | 1     | 2021/10/16 11:29:51                                                                           |
| sAMAccountName         | DirectoryString      | 1     | Administrator                                                                                 |
| sAMAccountType         | Integer              | 1     | 805306368                                                                                     |
| userAccountControl     | Integer              | 1     | 512                                                                                           |
| uSNChanged             | Integer8             | 1     | 0x31F3                                                                                        |
| uSNCreated             | Integer8             | 1     | 0x2004                                                                                        |
| whenChanged            | GeneralizedTime      | 1     | 2021/10/16 11:42:05                                                                           |
| whenCreated            | GeneralizedTime      | 1     | 2021/10/16 11:24:44                                                                           |

## 12. nTSecurityDescriptor

nTSecurityDescriptor 属性的值是这个条目的 ACL。

如图所示，可以看到 nTSecurityDescriptor 属性的值的类型为 NTSecurityDescriptor，其值就是该条目的 ACL，是用 SDDL 语言描述的。



| Attribute              | Syntax               | Count | Value(s)                                                                            |
|------------------------|----------------------|-------|-------------------------------------------------------------------------------------|
| accountExpires         | Integer8             | 1     | 0x0                                                                                 |
| adminCount             | Integer              | 1     | 1                                                                                   |
| badPasswordTime        | Integer8             | 1     | 2021/10/16 11:29:34                                                                 |
| badPwdCount            | Integer              | 1     | 0                                                                                   |
| cn                     | DirectoryString      | 1     | Administrator                                                                       |
| codePage               | Integer              | 1     | 0                                                                                   |
| countryCode            | Integer              | 1     | 0                                                                                   |
| description            | DirectoryString      | 1     | 管理计算机(域)的内置帐户                                                                       |
| distinguishedName      | DN                   | 1     | CN=Administrator,CN=Users,DC=xie,DC=com                                             |
| dSCorePropagationData  | GeneralizedTime      | 4     | 2021/10/16 11:42:05;2021/10/16 11:42:05;2021/10/16 11:26:55;1601/1/2 1:12:16        |
| instanceType           | Integer              | 1     | 4                                                                                   |
| isCriticalSystemObject | Boolean              | 1     | TRUE                                                                                |
| lastLogoff             | Integer8             | 1     | 0x0                                                                                 |
| lastLogon              | Integer8             | 1     | 2021/10/26 22:12:01                                                                 |
| lastLogonTimestamp     | Integer8             | 1     | 2021/10/16 11:29:51                                                                 |
| logonCount             | Integer              | 1     | 20                                                                                  |
| logonHours             | OctetString          | 1     | 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 |
| memberOf               | DN                   | 5     | CN=Group Policy Creator Owners,CN=Users,DC=xie,DC=com;CN=Domain Admins,CN=Users     |
| name                   | DirectoryString      | 1     | Administrator                                                                       |
| nTSecurityDescriptor   | NTSecurityDescriptor | 1     | D:PAI(OA;RP;4c164200-20c0-11d0-a768-00aa006e0529;4828cc14-1437-45bc-9b07-ad6f01     |
| objectCategory         | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                                  |
| objectClass            | OID                  | 4     | top;person;organizationalPerson;user                                                |
| objectGUID             | OctetString          | 1     | {7B73B1AA-7830-48F7-9106-0BE3FA003BCE}                                              |
| objectSid              | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-500                                       |
| primaryGroupID         | Integer              | 1     | 513                                                                                 |
| pwdLastSet             | Integer8             | 1     | 2021/10/16 11:29:51                                                                 |
| sAMAccountName         | DirectoryString      | 1     | Administrator                                                                       |
| sAMAccountType         | Integer              | 1     | 805306368                                                                           |
| userAccountControl     | Integer              | 1     | 512                                                                                 |
| uSNChanged             | Integer8             | 1     | 0x31F3                                                                              |
| uSNCreated             | Integer8             | 1     | 0x2004                                                                              |
| whenChanged            | GeneralizedTime      | 1     | 2021/10/16 11:42:05                                                                 |
| whenCreated            | GeneralizedTime      | 1     | 2021/10/16 11:24:44                                                                 |

其实与用户 ACL 相关的还有一个属性，就是 defaultSecurityDescriptor 属性。defaultSecurityDescriptor 属性是实例默认的 ACL。但是在 administrator 条目这里没看到这个属性，我们可以在 Schema Directory Partition 里面查询一下即可看到 User 类还有一个 defaultSecurityDescriptor 属性。也就是说，当 User 类在实例化的时候，没有指定 ACL 的话，则 nTSecurityDescriptor 属性的值就为 CN=User,CN=Schema,CN=Configuration,DC=xie,DC=com 条目的 defaultSecurityDescriptor 属性的值。如果在实例化的过程中，指定了 ACL 的话，则 nTSecurityDescriptor 属性的值就为指定的 ACL。

### 13. objectClass

objectClass 属性代表该实例对象继承的类有哪些。

如图所示，是 administrator 对象的 objectClass 属性，OID 代表其值数据类型为 OID，4 代表其值有 4 个，而 top;person;organizationalPerson;user 就是其属性的值。

| Attribute              | Syntax               | Count | Value(s)                                                                        |
|------------------------|----------------------|-------|---------------------------------------------------------------------------------|
| accountExpires         | Integer8             | 1     | 0x0                                                                             |
| adminCount             | Integer              | 1     | 1                                                                               |
| badPasswordTime        | Integer8             | 1     | 2021/10/16 11:29:34                                                             |
| badPwdCount            | Integer              | 1     | 0                                                                               |
| cn                     | DirectoryString      | 1     | Administrator                                                                   |
| codePage               | Integer              | 1     | 0                                                                               |
| countryCode            | Integer              | 1     | 0                                                                               |
| description            | DirectoryString      | 1     | 管理计算机(域)的内置帐户                                                                   |
| distinguishedName      | DN                   | 1     | CN=Administrator,CN=Users,DC=xie,DC=com                                         |
| dSCorePropagationData  | GeneralizedTime      | 4     | 2021/10/16 11:42:05;2021/10/16 11:42:05;2021/10/16 11:26:55;1601/1/2 1:12:16    |
| instanceType           | Integer              | 1     | 4                                                                               |
| isCriticalSystemObject | Boolean              | 1     | TRUE                                                                            |
| lastLogoff             | Integer8             | 1     | 0x0                                                                             |
| lastLogon              | Integer8             | 1     | 2021/10/26 22:12:01                                                             |
| lastLogonTimestamp     | Integer8             | 1     | 2021/10/16 11:29:51                                                             |
| logonCount             | Integer              | 1     | 20                                                                              |
| logonHours             | OctetString          | 1     | 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 |
| memberOf               | DN                   | 5     | CN=Group Policy Creator Owners,CN=Users,DC=xie,DC=com;CN=Domain Admins,CN=User  |
| name                   | DirectoryString      | 1     | Administrator                                                                   |
| nTSecurityDescriptor   | NTSecurityDescriptor | 1     | D:PAI(OA;;RP;4c164200-20c0-11d0-a768-00aa006e0529;4828cc14-1437-45bc-9b07-ad6f0 |
| objectCategory         | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                              |
| objectClass            | OID                  | 4     | top;person;organizationalPerson;user                                            |
| objectGUID             | OctetString          | 1     | {7B73B1AA-7830-48F7-9106-0BE3FA003BCE}                                          |
| objectSid              | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-500                                   |
| primaryGroupID         | Integer              | 1     | 513                                                                             |
| pwdLastSet             | Integer8             | 1     | 2021/10/16 11:29:51                                                             |
| sAMAccountName         | DirectoryString      | 1     | Administrator                                                                   |
| sAMAccountType         | Integer              | 1     | 805306368                                                                       |
| userAccountControl     | Integer              | 1     | 512                                                                             |
| uSNChanged             | Integer8             | 1     | 0x31F3                                                                          |
| uSNCreated             | Integer8             | 1     | 0x2004                                                                          |
| whenChanged            | GeneralizedTime      | 1     | 2021/10/16 11:42:05                                                             |
| whenCreated            | GeneralizedTime      | 1     | 2021/10/16 11:24:44                                                             |

而 objectClass 这个属性在 Schema Directory Partition 里面就有定义，如图所示：

Path: CN=Object-Class,CN=Schema,CN=Configuration,DC=xie,DC=com,10.211.55.4 [DC.xie.com]

| Attribute                  | Syntax               | Count | Value(s)                                                                                                |
|----------------------------|----------------------|-------|---------------------------------------------------------------------------------------------------------|
| adminDescription           | DirectoryString      | 1     | Object-Class                                                                                            |
| adminDisplayName           | DirectoryString      | 1     | Object-Class                                                                                            |
| attributeID                | OID                  | 1     | 2.5.4.0                                                                                                 |
| attributeSecurityGUID      | ObjectString         | 1     | {F48D0154-B0F8-11D1-8702-00C04FB96050}                                                                  |
| attributeSyntax            | OID                  | 1     | 2.5.5.2                                                                                                 |
| cn                         | DirectoryString      | 1     | Object-Class                                                                                            |
| distinguishedName          | DN                   | 1     | CN=Object-Class,CN=Schema,CN=Configuration,DC=xie,DC=com                                                |
| dSCorePropagationData      | GeneralizedTime      | 1     | 1601/1/1 7:00:00                                                                                        |
| instanceType               | Integer              | 1     | 4                                                                                                       |
| isMemberOfPartialAttrib... | Boolean              | 1     | TRUE                                                                                                    |
| isSingleValued             | Boolean              | 1     | FALSE                                                                                                   |
| LDAPDisplayName            | DirectoryString      | 1     | Object-Class                                                                                            |
| name                       | DirectoryString      | 1     | Object-Class                                                                                            |
| ntSecurityDescriptor       | NTSecurityDescriptor | 1     | D:AI(A;CIIID;LCRPLORC;;;AU)(A;CIIID;CCLCSWRPWPLOCRRCWDWO;;;SA)(A;CIIID;CCDCLCSWRPWPOTLOCRSRRCWDOO;;;SY) |
| objectCategory             | DN                   | 1     | CN=Attribute-Schema,CN=Schema,CN=Configuration,DC=xie,DC=com                                            |
| objectClass                | OID                  | 2     | top;attributeSchema                                                                                     |
| objectGUID                 | ObjectString         | 1     | {7B73B1AA-7830-48F7-9106-0BE3FA003BCE}                                                                  |
| objectSyntax               | Integer              | 1     | 6                                                                                                       |
| schemaFlagsEx              | Integer              | 1     | 1                                                                                                       |
| schemaIDGUID               | ObjectString         | 1     | {BF9679E5-0DE6-11D0-A285-00AA003049E2}                                                                  |
| searchFlags                | Enumeration          | 1     | 9                                                                                                       |
| showInAdvancedViewO...     | Boolean              | 1     | TRUE                                                                                                    |
| systemFlags                | Integer              | 1     | 18                                                                                                      |
| systemOnly                 | Boolean              | 1     | TRUE                                                                                                    |
| uSNChanged                 | Integer8             | 1     | 0x36E                                                                                                   |
| uSNCreated                 | Integer8             | 1     | 0x36E                                                                                                   |
| whenChanged                | GeneralizedTime      | 1     | 2013/5/22 0:44:35                                                                                       |
| whenCreated                | GeneralizedTime      | 1     | 2013/5/22 0:44:35                                                                                       |

如图所示，通过定义的语法查表可以得知其值的数据类型为 String(Object-Identifier)，也就是我们看到的 OID。

|                           |         |   |           |                       |
|---------------------------|---------|---|-----------|-----------------------|
| String(Object-Identifier) | 2.5.5.2 | 6 | \x5550502 | The object identifier |
|---------------------------|---------|---|-----------|-----------------------|

## 14. objectGUID

objectGUID 属性就是对象的 GUID 属性的值。

如图所示，可以看到 administrator 的 objectGUID 属性为{7B73B1AA-7830-48F7-9106-0BE3FA003BCE}。

| Attribute              | Syntax               | Count | Value(s)                                                                |
|------------------------|----------------------|-------|-------------------------------------------------------------------------|
| accountExpires         | Integer8             | 1     | 0x0                                                                     |
| adminCount             | Integer              | 1     | 1                                                                       |
| badPasswordTime        | Integer8             | 1     | 2021/10/16 11:29:34                                                     |
| badPwdCount            | Integer              | 1     | 0                                                                       |
| cn                     | DirectoryString      | 1     | Administrator                                                           |
| codePage               | Integer              | 1     | 0                                                                       |
| countryCode            | Integer              | 1     | 0                                                                       |
| description            | DirectoryString      | 1     | 管理计算机(域)的内置帐户                                                           |
| distinguishedName      | DN                   | 1     | CN=Administrator,CN=Users,DC=xie,DC=com                                 |
| dSCorePropagationData  | GeneralizedTime      | 4     | 2021/10/16 11:42:05;2021/10/16 11:42:05;2021/10/16 11:26:55;1601/1/2 1: |
| instanceType           | Integer              | 1     | 4                                                                       |
| isCriticalSystemObject | Boolean              | 1     | TRUE                                                                    |
| lastLogoff             | Integer8             | 1     | 0x0                                                                     |
| lastLogon              | Integer8             | 1     | 2021/10/26 22:12:01                                                     |
| lastLogonTimestamp     | Integer8             | 1     | 2021/10/16 11:29:51                                                     |
| logonCount             | Integer              | 1     | 20                                                                      |
| logonHours             | OctetString          | 1     | 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 :       |
| memberOf               | DN                   | 5     | CN=Group Policy Creator Owners,CN=Users,DC=xie,DC=com;CN=Domain Adm     |
| name                   | DirectoryString      | 1     | Administrator                                                           |
| nTSecurityDescriptor   | NTSecurityDescriptor | 1     | D:PAI(OA;;RP;4c164200-20c0-11d0-a768-00aa006e0529;4828cc14-1437-45b     |
| objectCategory         | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                      |
| objectClass            | OID                  | 4     | top:person:organizationalPerson:user                                    |
| objectGUID             | OctetString          | 1     | {7B73B1AA-7830-48F7-9106-0BE3FA0038CE}                                  |
| objectSid              | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-500                           |
| primaryGroupID         | Integer              | 1     | 513                                                                     |
| pwdLastSet             | Integer8             | 1     | 2021/10/16 11:29:51                                                     |
| sAMAccountName         | DirectoryString      | 1     | Administrator                                                           |
| sAMAccountType         | Integer              | 1     | 805306368                                                               |
| userAccountControl     | Integer              | 1     | 512                                                                     |
| uSNChanged             | Integer8             | 1     | 0x31F3                                                                  |
| uSNCreated             | Integer8             | 1     | 0x2004                                                                  |
| whenChanged            | GeneralizedTime      | 1     | 2021/10/16 11:42:05                                                     |
| whenCreated            | GeneralizedTime      | 1     | 2021/10/16 11:24:44                                                     |

## 15. objectSid

objectSid 属性代表该条目 SID 的值，域管理员 administrator 的 SID 默认为域 SID+500。

如图所示，可以看到 administrator 的 objectSid 属性 S-1-5-21-域 SID-500。







| Attribute              | Syntax               | Count | Value(s)                                                                     |
|------------------------|----------------------|-------|------------------------------------------------------------------------------|
| accountExpires         | Integer8             | 1     | 0x0                                                                          |
| adminCount             | Integer              | 1     | 1                                                                            |
| badPasswordTime        | Integer8             | 1     | 2021/10/16 11:29:34                                                          |
| badPwdCount            | Integer              | 1     | 0                                                                            |
| cn                     | DirectoryString      | 1     | Administrator                                                                |
| codePage               | Integer              | 1     | 0                                                                            |
| countryCode            | Integer              | 1     | 0                                                                            |
| description            | DirectoryString      | 1     | 管理计算机(或)的内置帐户                                                                |
| distinguishedName      | DN                   | 1     | CN=Administrator,CN=Users,DC=xie,DC=com                                      |
| dSCorePropagationData  | GeneralizedTime      | 4     | 2021/10/16 11:42:05;2021/10/16 11:42:05;2021/10/16 11:26:55;1601/1/2 1:12:16 |
| instanceType           | Integer              | 1     | 4                                                                            |
| isCriticalSystemObject | Boolean              | 1     | TRUE                                                                         |
| lastLogoff             | Integer8             | 1     | 0x0                                                                          |
| lastLogon              | Integer8             | 1     | 2021/10/26 22:12:01                                                          |
| lastLogonTimestamp     | Integer8             | 1     | 2021/10/16 11:29:51                                                          |
| logonCount             | Integer              | 1     | 20                                                                           |
| logonHours             | OctetString          | 1     | 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 ;        |
| memberOf               | DN                   | 5     | CN=Group Policy Creator Owners,CN=Users,DC=xie,DC=com;CN=Domain Admins,CN=   |
| name                   | DirectoryString      | 1     | Administrator                                                                |
| ntSecurityDescriptor   | NTSecurityDescriptor | 1     | D:PAI(OA;;RP;4c164200-20c0-11d0-a768-00aa006e0529;4828cc14-1437-45bc-9b07-ac |
| objectCategory         | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                           |
| objectClass            | OID                  | 4     | top;person;organizationalPerson;user                                         |
| objectGUID             | OctetString          | 1     | {7B73B1AA-7830-48F7-9106-0BE3FA003BCE}                                       |
| objectSid              | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-500                                |
| primaryGroupID         | Integer              | 1     | 513                                                                          |
| pwdLastSet             | Integer8             | 1     | 2021/11/8 15:11:28                                                           |
| sAMAccountName         | DirectoryString      | 1     | Administrator                                                                |
| sAMAccountType         | Integer              | 1     | 805306368                                                                    |
| userAccountControl     | Integer              | 1     | 512                                                                          |
| uSNChanged             | Integer8             | 1     | 0x50BD                                                                       |
| uSNCreated             | Integer8             | 1     | 0x2004                                                                       |
| whenChanged            | GeneralizedTime      | 1     | 2021/11/8 15:11:28                                                           |
| whenCreated            | GeneralizedTime      | 1     | 2021/10/16 11:24:44                                                          |

## 18. whenCreated

whenCreated 属性就是代表该对象被创建的时间。

如图所示，可以看到 administrator 的 whenCreated 属性为 2021/10/16 11:24:44。

| Attribute              | Syntax               | Count | Value(s)                                                                        |
|------------------------|----------------------|-------|---------------------------------------------------------------------------------|
| accountExpires         | Integer8             | 1     | 0x0                                                                             |
| adminCount             | Integer              | 1     | 1                                                                               |
| badPasswordTime        | Integer8             | 1     | 2021/10/16 11:29:34                                                             |
| badPwdCount            | Integer              | 1     | 0                                                                               |
| cn                     | DirectoryString      | 1     | Administrator                                                                   |
| codePage               | Integer              | 1     | 0                                                                               |
| countryCode            | Integer              | 1     | 0                                                                               |
| description            | DirectoryString      | 1     | 管理计算机(域)的内置帐户                                                                   |
| distinguishedName      | DN                   | 1     | CN=Administrator,CN=Users,DC=xie,DC=com                                         |
| dSCorePropagationData  | GeneralizedTime      | 4     | 2021/10/16 11:42:05;2021/10/16 11:42:05;2021/10/16 11:26:55;1601/1/2 1:12:16    |
| instanceType           | Integer              | 1     | 4                                                                               |
| isCriticalSystemObject | Boolean              | 1     | TRUE                                                                            |
| lastLogoff             | Integer8             | 1     | 0x0                                                                             |
| lastLogon              | Integer8             | 1     | 2021/10/26 22:12:01                                                             |
| lastLogonTimestamp     | Integer8             | 1     | 2021/10/16 11:29:51                                                             |
| logonCount             | Integer              | 1     | 20                                                                              |
| logonHours             | OctetString          | 1     | 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 255 |
| memberOf               | DN                   | 5     | CN=Group Policy Creator Owners,CN=Users,DC=xie,DC=com;CN=Domain Admins,CN=...   |
| name                   | DirectoryString      | 1     | Administrator                                                                   |
| ntSecurityDescriptor   | NTSecurityDescriptor | 1     | D:PAI(OA;;RP;4c164200-20c0-11d0-a768-00aa006e0529;4828cc14-1437-45bc-9b07...    |
| objectCategory         | DN                   | 1     | CN=Person,CN=Schema,CN=Configuration,DC=xie,DC=com                              |
| objectClass            | OID                  | 4     | top;person;organizationalPerson;user                                            |
| objectGUID             | OctetString          | 1     | {7B73B1AA-7830-48F7-9106-0BE3FA003BCE}                                          |
| objectSid              | Sid                  | 1     | S-1-5-21-1313979556-3624129433-4055459191-500                                   |
| primaryGroupID         | Integer              | 1     | 513                                                                             |
| pwdLastSet             | Integer8             | 1     | 2021/11/8 15:11:28                                                              |
| sAMAccountName         | DirectoryString      | 1     | Administrator                                                                   |
| sAMAccountType         | Integer              | 1     | 805306368                                                                       |
| userAccountControl     | Integer              | 1     | 512                                                                             |
| uSNChanged             | Integer8             | 1     | 0x50BD                                                                          |
| uSNCreated             | Integer8             | 1     | 0x2004                                                                          |
| whenChanged            | GeneralizedTime      | 1     | 2021/11/8 15:11:28                                                              |
| whenCreated            | GeneralizedTime      | 1     | 2021/10/16 11:24:44                                                             |

非常感谢您读到现在，由于作者的水平有限，编写时间仓促，文章中难免会出现一些错误或者描述不准确的地方，恳请各位师傅们批评指正。如果你想一起学习 AD 域安全攻防的话，可以加入下面的知识星球一起学习交流。



谢公子

我加入了这个星球，邀你一起学习



## 域渗透攻防指南

星主：谢公子

60+

成员数量

30+

内容数量

306

运营天数

书籍  《域渗透攻防指南》官方知识星球。  
星球讨论与微软AD域相关攻防技术，大家互相交流互相学习，共同进步成长。

现价：¥199

微信扫码加入星球



参考:

<https://docs.microsoft.com/en-us/windows/win32/adschema/active-directory-schema-site>

<https://docs.microsoft.com/en-us/windows/win32/ad/naming-contexts-and-partitions>

[https://docs.microsoft.com/en-us/openspecs/windows\\_protocols/ms-adts/f0f1e85a-3ac2-466a-87bf-2a8a1d18e4b8](https://docs.microsoft.com/en-us/openspecs/windows_protocols/ms-adts/f0f1e85a-3ac2-466a-87bf-2a8a1d18e4b8)

<https://docs.microsoft.com/en-us/windows/win32/adschema/classes>

<https://docs.microsoft.com/en-us/windows/win32/adschema/attributes>

[https://docs.microsoft.com/zh-cn/previous-versions/windows/it-pro/windows-server-2003/cc773309\(v=ws.10\)](https://docs.microsoft.com/zh-cn/previous-versions/windows/it-pro/windows-server-2003/cc773309(v=ws.10))

